

Time-varying Time Preferences*

M. Steven Holloway[†]

Michigan Technological University

Michael A. Kuhn[‡]

University of Oregon and CEGA

Connor Thomas Wiegand[§]

University of Oregon

September 15, 2026

Abstract

Preference reversals in intertemporal choice have long been considered hallmarks of “present-biased” discounting. While a nascent literature posits that they could instead result from time-varying discounting, the evidence comes from $T = 2$ longitudinal studies that cannot identify whether time-variance is noise or systematic change. We implement a longitudinal discounting study with seven elicitations over a 12-week period, and find that subjects exhibit increasing discounting over the course of the study, and that time-variance in discounting explains roughly 72% of time-inconsistent choices in our data. We introduce a new discounting model, the Nested Exponential discount function, which permits any feasible configuration of Halevy (2015)’s three properties of time preference: time-invariance, time-consistency, and stationarity. The function nests both exponential discounting and present-biased discounting within its parameter space, enabling data-driven property configurations at both the aggregate and subject levels.

JEL Classification: D15, D90, C90

Keywords: discounting, dynamic inconsistency, time preference, experiment

*IRB approval for this research was obtained from the University of Oregon, Study ID: 00000768.

[†]College of Business, 1400 Townsend Drive, Houghton, MI 49931, USA. E-mail: holloway@mtu.edu.
Web: www.mstevenholloway.page/home

[‡]Department of Economics, 1285 University of Oregon, Eugene, OR 97403, USA. E-mail: mkuhn@uoregon.edu. Web: www.makuhn.net

[§]Department of Economics, 1285 University of Oregon, Eugene, OR 97403, USA. E-mail: [cwiegand@uoregon.edu](mailto:cwie-gand@uoregon.edu). Web: ctwie.me

1 Introduction

A substantial body of evidence documents that economic preferences can be unstable. Beyond the large and still growing literature showing causal effects of shocks on preferences, decision-makers may learn about their preferences over time (either through feedback (Charness et al., 2023), the realization of uncertainty (Carrera et al., 2022), or deliberation (Gabaix and Laibson, 2022)), as they mature (Andreoni et al., 2019), or “choose” preferences to better suit their environment (Bernheim et al., 2021). Preference instability, shocks, and drift are problematic for researchers looking to estimate time preferences. Consider the framework of Read and van Leeuwen (1998) where healthy eating is taken as having an immediate opportunity cost in exchange for future health benefits: in advance of consumption an individual selects a healthy snack for their future self, but revises their choice to an unhealthy snack when the time for consumption arrives. Is this preference reversal best explained by a fixed feature of that individual’s discount function, learning about preferences, or some other unobserved environmental factor?

Halevy (2015) describes three properties of time preferences: consistency, stationarity, and time-invariance. The healthy-to-unhealthy snack preference reversal example above shows inconsistent behavior: holding fixed the *calendar dates* of the consequences of an intertemporal choice, time-inconsistency is when the date the choice is made affects the outcome. This is the key behavior that has motivated much of the past 25 years of literature in behavioral economics surrounding why people don’t stick to their plans. However, directly observing inconsistency requires a longitudinal setting, which is often difficult to engineer. Thus, many researchers looking to measure inconsistency instead measure the closely related property of stationarity. Sticking with the snack example, simultaneously choosing an unhealthy snack for now and a healthy snack for the future is an example of non-stationary time preference: holding fixed the date a choice is made, non-stationarity happens when the *relative dates* of the choice consequences affect the outcome.

Non-stationarity is an important behavior to study in its own right, but under what conditions does it also inform researchers about inconsistency? Halevy (2015) shows that

they are equivalent only when time-invariance also holds. Again using the snack example, choosing an unhealthy snack now for now, and then choosing a healthy snack tomorrow for tomorrow is an example of time-varying time preference. Essentially, non-stationarity and inconsistency are equivalent if and only if individuals exhibit stable choices in identical intertemporal choice situations over time.

Any two of these properties imply the third; if a behavioral economist theorizes that a decision-maker is time-inconsistent, she necessarily assumes that the decision-maker is either non-stationary or time-variant (or both). Common models of discounting in behavioral economics pair inconsistency with non-stationarity, while leaving time-invariance intact. However, data show this is only one of many choice patterns that people exhibit. In Halevy (2015), at most 68%, and as few as 44% of subjects exhibited either all three properties or only time-invariance. Thus, between 32% and 56% of the sample exhibited preferences that cannot be explained with current models. In a closely-related paper, Janssens et al. (2017) finds that while 43% of subjects in their study violate consistency, only 24% violate *both* consistency and stationarity. Similarly, Harrison et al. (2025) finds that time-variance explains why such a large fraction of their experimental sample is inconsistent even though they display stationary behavior in their first of their two surveys. Time-variance is an important reason why subjects in all three studies exhibit inconsistent behavior.

In this paper we present both a twelve-week, seven-survey, longitudinal experiment that offers much higher resolution on time-variance than past work, and a structural model that flexibly accounts for time-variance in intertemporal choices. While the two-period studies in Halevy (2015), Janssens et al. (2017), and Harrison et al. (2025) allow for an impact of time-variance on experimental measures, they do not give the researcher much power to consider the nature of time-variance and model it. Is time-variance in the canonical experimental context for intertemporal choices noise or systematic? Recent work in behavioral economics establishes numerous potential pathways for systematic change in time preference: a likely non-exhaustive list includes preference uncertainty (Carrera et al., 2022), deliberation (Gabaix and Laibson, 2022) and cooling off (Imas et al., 2022), aging (An-

dreoni et al., 2019), adaptation (Bernheim et al., 2021), reductions in cognitive uncertainty (Enke et al., 2025), and correlated economic factors (Carvalho et al., 2016). Answering this question determines the scope of the time-variance problem. If time-variance is idiosyncratic noise, then what appears to be time-variance or inconsistency may instead arise from decision error. Failing to account for this would lead to individuals being incorrectly classified as certain types of discounters, but would not impact aggregate estimates of how a population discounts (besides decreasing precision). On the other hand, if time-variance is a systematic process the problem is more pernicious.

What is the risk associated with mistaking inconsistency that corresponds to a stationarity violation for inconsistency that corresponds to time-variance? The behavioral literature on “present-bias” (Laibson, 1997; O’Donoghue and Rabin, 1999) and associated policy interventions (e.g. commitment devices) are predicated on inconsistency as a harmful effect of the opportunity for immediate gratification. If choices can be structured to avoid that opportunity, then the decision-maker can implement their stable long-run preference for saving more, eating healthier, etc. Inconsistent choices associated with time-variance do not have the same interpretation, and such interventions may decrease welfare. This risk can be mitigated simply by measuring time-variance. Take the following example: consider a decision-maker that exhibits time-varying preferences such that his intertemporal allocations in identical choice scenarios are getting less patient as time passes. First, assume his preferences are stationary. If a researcher estimates his quasi-hyperbolic (β - δ) discount function via an observed violation of consistency, she will conclude that he is present-biased ($\beta < 1$). Now instead, assume his preferences are consistent. If his quasi-hyperbolic discount function is estimated via an observed violation of stationarity, the researcher will conclude that he is future-biased ($\beta > 1$) instead. If the researcher also measures the violation of time-invariance, she will learn in the first (second) case that the decision-makers degree of inconsistency (non-stationarity) is exactly equal to their degree of time-variance. Suitably adjusted for the observed time-variance, $\beta = 1$ in both cases.

Section 2 gives additional background on the three properties of time preferences, as

illustrated by Halevy (2015), and why time-variance is a problem for identification in time-invariant models. Sections 3-5 present our longitudinal experiment. Over the course of 12 weeks subjects take part in seven surveys each featuring 6- 10 incentivized intertemporal choices. For every choice, this design gives us six opportunities to observe inconsistency (one for each pair of adjacent surveys), six opportunities to observe non-stationarity (within each survey except the last) and twenty one opportunities to observe time-variance (one for all pairs of surveys, though we focus on the six adjacent pairs). We first look at our experimental data through the lens of traditional time-invariant discounting, and estimate parameters for the quasi-hyperbolic β - δ discount function. We identify small but statistically significant present bias at the aggregate level, and the majority of subjects are classified as present-biased at the individual level. We then shift to analyses that allow for time-varying preferences. We identify a significant time trend in the aggregate discount factor consistent with *decreasing patience* (increasing discounting) over the period of the study. This trend stems from subject-level changes across time, not sample composition, suggesting systematic rather than idiosyncratic time-variance. With time-variance accounted for, we can measure distinct inconsistent and non-stationary behavior at the subject-level; we find that “inconsistency present-bias” is larger and more widespread than “non-stationary present bias.” The prevalence of individual violations of all three time preference properties leads us to classify subjects as time-variant at the upper end of the range Halevy (2015) suggests. Overall, time-variance explains 72% of the inconsistency in our sample.

Our interpretation of decreasing patience during our study is not that it represents a general trend in intertemporal choice behavior over time that would extrapolate to substantially longer time horizons. Instead, we think it more likely that we are recovering either local drift in preferences specific to our population and time period, or an artifact of the repeated experimental choices themselves like those mentioned earlier: resolving preference uncertainty, further deliberation about the choice options, or some other pathway involving subjects learning about the choices they want to make. In this light, we view our empirical contribution to the literature as demonstrating systematic time-variance *in a canonical ex-*

perimental context. Our choice intervals of every two weeks over a 12-week period were designed to inform this context, rather than long-run aging processes over the course of years, or immediate responses to feedback over the course of hours.

In Section 6 we discuss how discount functions reflect the time preference properties, and present a new model – the “nested exponential” discount function– that is general with respect to the properties defined by Halevy (2015). Nested Exponential Discounting (NED) can satisfy any of the five possible combinations of properties depending on its parameter values: all three properties can fail, all three can hold, or any single property can hold on its own. While this four-parameter model is of course not as tractable as the workhorse β - δ model, we develop the intuition of the model as a discrete extension of fast Weibull discounting (Jamison and Jamison, 2011), one of the less-common behavioral models that permits inconsistency and non-stationarity. Thus, the model retains a “present-bias” parameter similar to existing models.

In Section 7 we structurally estimate the NED function at both the aggregate and individual levels. We directly test the estimated parameters for adherence with time-invariance, consistency, and stationarity, and engage in model selection exercises where we horse-race restricted and unrestricted versions of the NED. The unrestricted model fits the data best, even when penalized for its extra parameters. However the next best performing model is the version that is constrained to be time-invariant only. In other words, classic “present-bias” where inconsistency and non-stationarity go together does a decent job of explaining aggregate behavior so long as the researcher uses a discount function with some true hyperbolicity, rather than the quasi-hyperbolic β - δ model.¹ At the individual level, we again find that the unrestricted NED model best-fits a plurality of our subjects, but with a substantial group of invariant-only discounters. The remaining individuals are mostly categorized as stationary-only discounters, with a negligible share of exponential and consistent-only discounters. Overall, we assign 63% of the sample to time-varying models of discounting, suggesting that understanding, measuring, and modeling preference shifts over time is

¹Hyperbolicity is a key feature of our data; among one-parameter models, the single-parameter hyperbolic discount function outperforms exponential discounting.

crucial for accurately estimating time preferences.

2 Background

We adopt notation from Halevy (2015) to describe the stationarity, consistency, and time-invariance properties of time preferences. Consider an agent in calendar time period $\tau \geq 0$ evaluating bundles (ε, t) and (ψ, t') , which deliver rewards ε, ψ at time periods $t' \geq t \geq \tau$. We assume that the agent has complete and transitive preferences over all such bundles, according to a preference ordering that may depend on τ , $\succsim_\tau$. Under classical consumer theory assumptions, including additive separability of utility across time periods, this preference relationship is represented by a *discount function*, $D(\tau, t)$, such that

$$(\varepsilon, t) \succsim_\tau (\psi, t') \iff D(\tau, t) \cdot \varepsilon \geq D(\tau, t') \cdot \psi.^2 \quad (1)$$

$D(\tau, t)$ is the agent's discount factor when evaluating, in period τ , a payoff in period t .

Definition 2.1. An agent's preferences satisfy stationarity if

$$D(\tau, t) \varepsilon \geq D(\tau, t') \psi \iff D(\tau, t + \Delta) \varepsilon \geq D(\tau, t' + \Delta) \psi \quad \text{for } \Delta \geq 0.$$

An agent whose preferences are *time stationary* will not exhibit preference reversals when the payoff time of all bundles is shifted by the same number of periods (Δ), all else equal.

Definition 2.2. An agent's preferences satisfy consistency if

$$D(\tau, t) \varepsilon \geq D(\tau, t') \psi \iff D(\tau + \Delta, t) \varepsilon \geq D(\tau + \Delta, t') \psi \quad \text{for } 0 \leq \Delta \leq t - \tau.$$

Agents with *time consistent* preferences will not exhibit preference reversals when the evaluation period of all bundles is shifted by the same number of periods (Δ), all else equal.

²Without loss of generality, we replace $u(\varepsilon)$ and $u(\psi)$ with ε and ψ , under the assumption an instantaneous utility function representation exists.

Definition 2.3. An agent’s preferences satisfy time invariance if

$$D(\tau, t) \varepsilon \geq D(\tau, t') \psi \iff D(\tau + \Delta, t + \Delta) \varepsilon \geq D(\tau + \Delta, t' + \Delta) \psi .$$

Agents whose preferences are *invariant* will not exhibit preference reversals when the evaluation period and payoff periods for all bundles are shifted by the same amount (Δ), all else equal.

We use TICS (time-invariance, consistency, stationarity) to refer jointly to these three properties of time preferences. Halevy (2015) proves the following proposition, linking these three properties together:

Proposition I. *The TICS properties form a binary. That is, if a preference relation satisfies any two of the properties, then it must imply the third.*

Thus, the TICS properties generate five groups that any agent’s preferences may fall into: all hold, time-invariant only, stationary only, consistent only, none hold. The consequences of this binary relationship are not well-appreciated in the literature. Economists who wish to model inconsistent agents, for instance, must also depart from at least one of stationarity or time-invariance. The former approach is typical; the hyperbolic and β - δ discount functions satisfy only time-invariance, for example. There exists only a nascent literature on discount functions that do not satisfy time-invariance (Strulik, 2021), and on observed time-variant behavior (Halevy (2015); Janssens et al. (2017); DeJarnette (2020); Imas et al. (2022); Brownback et al. (2025); Harrison et al. (2025)).

Before proceeding, there are two terms that we will define for use in this paper that are convenient short-hands for phenomena we observe, but come with a lot of baggage in the literature: *(im)patience* and *present/future-bias*. In some papers these terms are closely linked (e.g. Prelec (2004), where present-bias is considered “decreasing impatience”). However, it will be useful for us to keep them separate in order to use terms like increasing and decreasing patience to describe time-variance. Specifically, a subject in our study exhibits decreasing (increasing) patience if their discount factor for a delayed reward

is decreasing (increasing) as time passes:

$$D(\tau, t) > D(\tau + \Delta, t + \Delta) \quad \text{for } t > \tau \text{ and } \Delta > 0 \quad . \quad (2)$$

Present (future) bias refers to the phenomenon where an agent’s relative preference for a sooner reward vs. a later reward is decreasing (increasing) in how far those rewards are from the *initial* evaluation period. Present bias indicates either a consistency ($C > 0$, $S = 0$) violation, a stationarity violation ($S > 0$, $C = 0$), or both ($C, S > 0$) of the form

$$\frac{D(\tau, t)}{D(\tau, t')} > \frac{D(\tau + C, t + S)}{D(\tau + C, t' + S)} \quad \text{for } t' \geq t, \quad S \geq 0 \text{ and } 0 \leq C \leq t - \tau + S \quad . \quad (3)$$

Often present bias is modeled discontinuously, with extra discounting applied to all non-immediate rewards in the workhorse quasi-hyperbolic discounting model (Laibson, 1997; O’Donoghue and Rabin, 1999). However, present bias is also a feature of the hyperbolic discounting model widely used outside economics. There is also debate over whether “bias” is appropriate to describe an empirical phenomena that could stem from a variety of root causes (Bernheim and Taubinsky, 2018). In this paper we use “present bias” and “future bias” simply to classify the direction of violations of stationarity and consistency.

2.1 Measuring discounting properties

Consider a decision maker (DM) faced with a choice evaluated at time $\tau = 0$ between an immediate reward at $t = 0$ and a reward on future date $t_1 > 0$. Let a denote a measure of how much their choice favors the reward at the later date, t_1 . That is, higher values of a result from less discounting of (a higher discount factor for) the delayed reward. In the same evaluation period ($\tau = 0$), the DM also faces a choice between utility in two future time periods, $t_2 > 0$ and $t_2 + t_1$. Let b denote the corresponding measure of how much their choice favors the later reward in that scenario. After time passes, such that calendar time is now $\tau' = t_2$, the DM faces a choice similar to the original: an immediate reward at $t = \tau'$ or a reward at future date $t_2 + t_1$. Once again, let c denote the measure of how

much their choice favors the later reward. We call the collection of these three choices a “decision triangle,” visualized in Figure 1. Using the values associated with a triangle, $a - c$ is a measure of time-variance; how much a choice over fixed relative dates changes over time. $b - c$ measures inconsistency; how much a choice over fixed calendar dates changes over time. Finally, $b - a$ measures non-stationarity; how much a choice made on a fixed date changes in response to front-end delay. Combining them, we have that

$$\underbrace{a - c}_{\text{Time Variance}} = \underbrace{a - b}_{\text{Non-Stationarity}} + \underbrace{b - c}_{\text{Inconsistency}}. \quad (4)$$

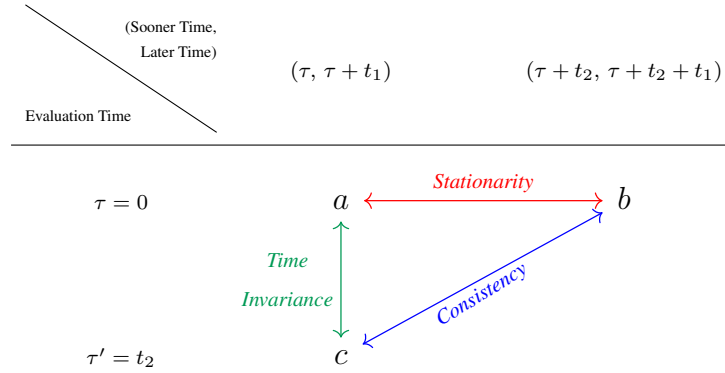


Figure 1: Time preference property decision triangle

With this framework, we can illustrate the empirical issues that arise when a researcher does not account for time-variance. Suppose a researcher wishes to estimate parameters of the β - δ model, and they set up an experiment to measure the DM’s time-consistency. Assume the DM exhibits time-varying choices in the form of decreasing patience, leading to $a - c < 0$. Assume also that the DM *would* exhibit stationary choices if they were measured, such that $b - a = 0$. In this case, the DM must then exhibit time-inconsistent choices such that $b - c = a - c < 0$. Thus, the researcher will conclude that the DM is present-biased and estimate $\beta < 1$. Now suppose instead that the researcher sets up an experiment to measure the DM’s stationarity. Assume the same direction of time-variance

in choices exists as the previous example, but also assume that the DM *would* exhibit time-consistent choices if they were measured, Since $b - c = 0$, it must be that the DM makes a non-stationary choice so that $a - b = a - c < 0$. Thus, the researcher will find the DM is future-biased, and estimate $\beta > 1$. Attribution of the agent’s true behavioral response to an immediate gratification opportunity in the face of identical, unobserved time-varying preferences can thus depend on research design.

3 Experimental Procedures

The details of our experimental design and analysis were pre-registered on AsPredicted.org, protocol #133180. We recruited subjects from introductory economics courses at the University of Oregon. A total of 178 subjects were invited to participate in the experiment, and 153 consented and completed the first survey. After initial recruitment, which occurred both in-person and over email, all communication with subjects was conducted online through the Qualtrics survey platform. While the frequent, short points of contact in our design make an online design preferable, because of the delayed and risky incentives we decided it was important to run the study in a sample where we could leverage our institutional reputation to establish trust, and where subjects would have in-person access to the study administrators in case of any problems or concerns. Participants were provided with a study email address, as well as the office location and phone number for the faculty supervisor.

The study featured seven surveys; one every two weeks for twelve weeks. Every subject that completed the first survey earned \$5, paid upon completion, and every subject that completed the final survey earned an additional \$35 upon completion. All payments were via instant digital transfer on a subject’s preferred platform.³ The first survey had detailed instructions on our protocol and the last featured an exit questionnaire. Both took subjects roughly 20 minutes to complete. Each intermediate survey featured only prize drawings, price lists, and a very brief questionnaire, each taking roughly 10 minutes to complete.

³Subjects chose from Venmo, PayPal, CashApp, and Zelle, and we pre-screened potential participants for their using one of these platforms to receive payment.

Within each survey, subjects faced multiple price lists (MPLs) that asked them to choose between two columns, where each column represented a prize-drawing for \$300 in a specified time period, and each row represented a choice between a ticket-time period bundle. Each ticket corresponded to an increment of 1 in 100,000 in odds within a drawing. The left column always offered tickets in a drawing that would take place earlier than the drawing in the right column. For all subjects, in every price list, the left column featured a sequence of 22 rows with 202 to 244 tickets by increments of two, whereas the right column featured a fixed 242 tickets in every row. This design is an adaptation of the multiple-lottery-list elicitation of Belot et al. (2025). See Appendix Figure A.1 for an example price list.

We denote surveys by the number of weeks from the start of the study that they occur (corresponding to t and τ in the discount functions). In weeks zero and two (the first and second surveys) we offered subjects six choices: choices between a drawing immediately and drawings in two, four, and eight weeks, and choices between a drawing in two weeks and drawings in four, six, and ten weeks. We refer to the time between drawings within a choice as the “choice delay” and the time until the first drawing within a choice as the “front-end delay.” When a subject responds to sequential surveys, we observe TICS decision triangles (one for each choice delay), allowing us to detect violations of any of three time preference properties. Starting in week four, we phased in longer choice delays of 16 and 32 weeks, at first only with the two-week front-end delay, but then in week six, without the front-end delay as well.⁴ Week 12 was the final week that involved subject choices, and we did not elicit choices with the front-end delay in this survey. Figure 2 outlines the price lists faced by subjects in each survey.

The goal of price list elicitation is to observe a single switching point between columns, such that the switch point approximately identifies at which row—as the value of one of the column’s rewards changes monotonically—a subject is indifferent between the bundle in the left column and the bundle in the right column. If the subject switches more than once, this confounds that identification. A single switching point in tickets was enforced by our

⁴In our regression models we will use delay-length fixed effects to flexibly account for these additional choices.

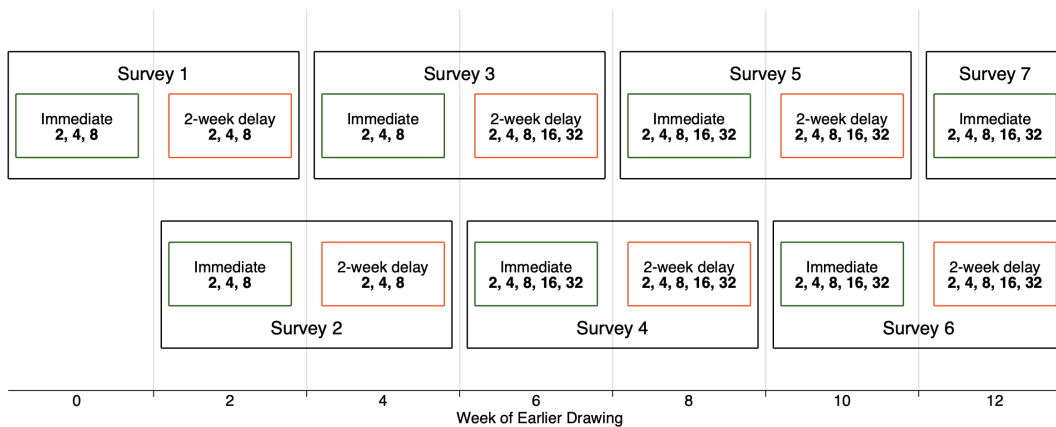


Figure 2: Study timeline

Notes: The green boxes represent price lists with immediate earlier drawings and the orange boxes represent price lists with earlier drawings in two weeks from the date of the survey. The numbers below these labels represent the choice delays of each price list administered –the number of weeks between the earlier and later drawings.

elicitation device, a bright yellow sliding bar that separated choice rows for which the subject preferred the earlier drawing from the choices for which subjects preferred the later drawing. Subjects were instructed to place the sliding bar *between* the last choice in which they preferred in the right column (the later prize drawing) and the first choice they preferred in the left column (the sooner prize drawing). Specifically, we instructed them to put every row where they prefer the later drawing above the bar, and every row where they prefer the earlier drawing below the bar. Writing on the bar itself expressed their switch point in words, which changed when the bar was dragged to a new location. Subjects could leave the bar at the very top –indicating they always prefer the earlier drawing– but they had to actively click and release it to do so.⁵ Subjects could not advance through a price list for at least six seconds in order to limit random or thoughtless choices.

All choices had a positive probability of being realized, but only one *per earlier-drawing date* was selected to count. For example, in week zero, with equal probability one of the price lists with an immediate earlier drawing was randomly selected, and then one of 22 rows was randomly selected, again with equal probability. If the subject picked

⁵We thank Antonia Krefeld-Schwalb for the idea, and the source code to help us build this tool.

the immediate drawing in that row, the drawing took place at the end of the week zero survey. If the subject picked the later drawing in that row, the drawing took place *at the beginning* of the survey in that week. The week zero price lists with a two-week front end delay were “saved for later”: when subjects returned in week two and made another set of choices featuring an earlier drawing on that date, the price list that counted for that earlier date was randomly selected from both the sets of week zero and week two choices.

Prize drawings were implemented with the selection of a random number from 1 to 100,000. If this number was less than or equal to the number of tickets a subject had in a drawing, they won. Winning triggered an email to a study administrator who would immediately transfer the \$300 prize. Three subjects won drawings, including one in the first survey. Many drawings took place after the surveys ended; in these cases, subjects received emails with a link to participate in the drawing with no data collection.

Before participating in incentivized price lists in week zero, subjects had to complete a number of comprehension tasks: 1) fill out an example list (however they wanted), 2) correctly use the yellow slider bar to illustrate a specified preference, 3) choose a random drawing number that would allow one hypothetical subject to win and another to lose based on their choices, and 4) correctly answer a question about the independence of choices across price lists. Each survey after week zero included a refresher question that subjects had to correctly answer prior to making their incentivized choices.⁶

The first survey was distributed by email at 7am on a Wednesday in the middle of the university term so that subjects would find the link in their inbox in the morning. All subsequent surveys were also distributed at 7am on Wednesday, covering the end of that term, finals week, and much of the subsequent term.⁷ Each survey was open for 26 hours,

⁶While the data are not the subject of the current paper, each survey concluded with a brief questionnaire on recent life events. Subjects were asked about financial and psychological shocks they may have experienced since the last survey. The same questions were asked every survey, except for one rotating question that assessed risk preferences, cognition, and other measures of time preferences not captured by our main elicitation. The final survey also included an exit questionnaire that asked subjects about their understanding of the survey, demographics and socioeconomic status, expectations about the future, subjective time horizon, patience, and impulsivity.

⁷The term length is 10 weeks, so we were unable to avoid an end of term somewhere in the study. Between having a survey in finals week or spring break, we selected finals week under the assumption that students

from 7am on Wednesday to 9am on Thursday. Subjects were allowed to miss one out of the seven surveys and were barred from further participation after their second missed survey.

3.1 Attrition

While a concern in any longitudinal study, attrition was a particular concern for this experiment given its length, number of contact points, and the potential for non-random attrition. If those that leave the study are less patient than those who remain –because of the delayed incentives– then the full sample may exhibit increasing-patience time-variance in the aggregate despite no individual level time-variance. Given resource constraints to offering overwhelming incentives and time constraints in hunting down missing individuals to complete a survey within a 26-hour window, we opted for an approach that 1) made completing each survey fast and easy, with multiple reminders and direct, individualized participation links (as opposed to requiring login information), 2) allowed us to carefully test for any selective attrition, and 3) allowed us to measure time-variance at the subject level.

Of the 153 subjects that completed the first survey, 97 (63%) completed the last survey. 79 (52%) of subjects completed every survey. This defines a “full” sample of 6,367 price lists from 153 subjects and a “balanced” sample of 4,345 price lists from 79 subjects. Throughout the results we present findings from both samples when it is concise to do so, focus on the balanced sample otherwise, presenting corresponding results from the full sample in the Appendix. None of the main results of the paper depends on the sample we use. We formally test for differential attrition in Section 5, and find no significant differences in the discount factor at any point in the study between those who are about to attrit and those who will remain (see Appendix Figure A.6).

One mitigating factor for concerns about attrition is our focus on subject-level results. We designed the study to observe the time preference properties exhibited by subjects and estimate subject-level parameters of discount functions. For all subjects that completed at least two subsequent surveys –141 in total, 92% of the initial sample– our experiment provides sufficient data to directly inform our research question.

would at least be on their computers, even if they were busy.

4 Time-invariant Behavior

We measure subject i 's implied discount factor in price list j in survey week w , d_{ijw} , as the midpoint of their switch interval divided by 242 (the fixed ticket allocation to the later drawing).⁸ For example, if a subject prefers 242 tickets later to 230 tickets earlier, but 232 tickets earlier to 242 tickets later, then $d = \frac{231}{242} = 0.955$.

Across all price lists the average d is 0.957, corresponding to a switch point between 230 and 232 tickets. For the balanced sample, the average d is 0.956, corresponding to the same switch point. These are significantly different from one ($p < 0.0001$ in both cases), consistent with a positive discounting over tickets, but not significantly different from each other ($p = 0.752$).⁹ Figure 3 shows that in general subjects discount more when the choice delay is longer, and when the earlier prize drawing is immediate. In fact, at every delay length, subjects discount the future more when the sooner draw is immediate, consistent with present-bias. In the full sample when there is no front-end delay, discounting is not decreasing in delay length from two to eight weeks. However, this is not true when we use the balanced sample and otherwise, discounting responds as expected to delay.

We regress d on choice delay length and an immediate indicator, and find that for each week the delay increases, d decreases by 0.001 on average in both the full and balanced samples ($p = 0.001$ in both cases). When the earlier lottery is immediate, d decreases by 0.003 (0.005 in the balanced sample) on average ($p = 0.007$ and $p = 0.001$, respectively). Given that empirical discount functions are often convex, we also estimate log-dependent variable models to allow for exponential decay. The coefficient estimates are very similar,

⁸When a subject never switches –always preferring the later draw– we assume a switch midpoint of 245. When a subject switches immediately –always preferring the sooner draw– we assume a switch midpoint of 201. In Section 4.1 we find that allowing for a censoring process at the endpoints instead has no impact on estimates of discounting.

⁹We run an OLS regression of d on an indicator for being in the balanced sample, with standard errors clustered at the subject level, and then perform two-tailed t -tests of equality with one for both groups.

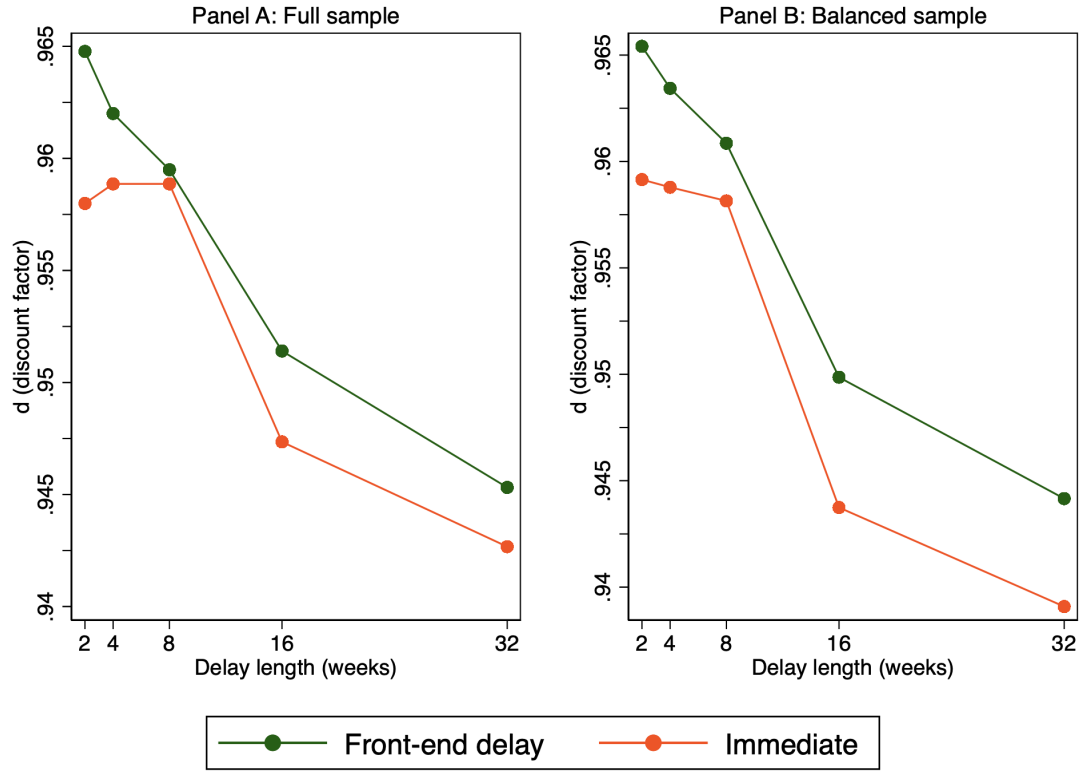


Figure 3: Empirical discount functions

Table 1: Discounting and price list features

Sample:	Full		Balanced	
	d	$\ln(d)$	d	$\ln(d)$
Dependent variable:	(1)	(2)	(3)	(4)
Delay (weeks - 8)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)
Immediate	-0.003*** (0.001)	-0.004*** (0.001)	-0.005*** (0.001)	-0.005*** (0.001)
Constant (Delay = 8 weeks, Immediate = 0)	0.960 (0.003)	-0.043 (0.003)	0.960 (0.004)	-0.042 (0.004)
Clusters	153	153	79	79
Observations	6,367	6,367	4,345	4,345

Notes: OLS regressions with standard errors clustered at the subject level. Delay is defined relative to eight weeks.

which we should expect with minimal misspecification bias and d close to one. Estimates are in Table 1. Subjects discount future rewards, do so more when the future is more distant, and exhibit small but significant present-bias.

4.1 Aggregate discounting estimates

A common application of time preference elicitation in the literature is to structurally estimate discounting parameters –typically an exponential δ parameter, and often a quasi-hyperbolic β parameter as well. Before proceeding with this exercise we illustrate how our data allow us to identify such parameters. The goal of our price list elicitation is to identify tight bounds on the set containing the indifference point between the two prize draws. When subjects never switch (always choose the later draw) or switch immediately (always choose the sooner draw), that set is only bounded below or above. 90% (91%) of choices in the balanced (full) sample are *interior* switch points, and only one subject failed to deliver a single interior switch point (they were in the balanced sample). As such, we feel comfortable using structural estimation strategies that assume a price list switch point identifies a point of *equality* between the sooner and later options (a “convex price list” in the terminology of Kuhn (2025)) for our main estimates. We will also show evidence for the robustness of our estimates to set-identification and binary-choice approaches.

Following our notation from Section 2 where the discount factor, $D(\tau, t)$, is a function of payoff period t and evaluation period τ , the quasi-hyperbolic form is $D(\tau, t) = \beta^{\mathbb{1}(t > \tau)} \cdot \delta^{t-\tau}$. For our sooner and later options within each price list, we call t^S and t^L the payoff dates of the respective lotteries.¹⁰ Assuming d identifies the point of indifference between the sooner and later draws in the price list, we have that

$$\beta^{\mathbb{1}(t^S > \tau)} \cdot \delta^{t^S - \tau} \cdot (X - 1) = \beta \cdot \delta^{t^L - \tau} \cdot 242 \Rightarrow d = \frac{\beta \cdot \delta^{t^L - \tau}}{\beta^{\mathbb{1}(t^S > \tau)} \cdot \delta^{t^S - \tau}} = \beta^{\mathbb{1}(t^S = \tau)} \cdot \delta^{t^L - t^S}, \quad (5)$$

where X is the number of sooner tickets in the switching row (and thus $d = \frac{(X-1)}{242}$). We

¹⁰Note that t^L is always greater than τ .

log both side of this equality to obtain the regression equation

$$\ln(d_{ijw}) = \mathbb{1}(t_j^S = \tau_j) \cdot \ln(\beta) + (t_j^L - t_j^S) \cdot \ln(\delta) + \varepsilon_{ijw}, \quad (6)$$

which we estimate with standard errors clustered at the subject level. Estimates of δ and β from this approach are shown in column (1) of Table 2. Column (2) shows the estimates from an interval regression (a generalized Tobit model), which uses only the known bounds of each switch interval, including one-sided bounds at the extremes. Column (3) shows estimates from a non-linear least squares (NLS) regression applied directly to equation 5. NLS will be our preferred technique for estimating the time-variant discount functions, which cannot be neatly linearized, so we include it here to show that it produces nearly identical estimates to other models. When price lists feature larger intervals than ours, or do not enforce a single switch point, researchers often take a discrete choice approach to the data at the choice-row level, using maximum likelihood to estimate the model parameters. We take this approach with a logistic choice model, with estimates in column (4).

Estimates for β and δ are nearly invariant to estimation method. Across the four models, our estimate of the annual discount rate ranges from 9% to 11%. This is consistent with the lower end of estimates from the time preference elicitation literature. For example Andersen et al. (2008) and Andersen et al. (2014) find estimates of 10% and 9% in nationally-representative samples of Danes, respectively. Andreoni and Sprenger (2012) estimates an annual rate of 37% using the Convex Time Budget technique with a US undergraduate population, and Kuhn et al. (2017) estimates a rate between 20% and 24% using the same technique among French undergraduates. Our estimates for β are all about 0.97, corresponding to present-bias that is roughly equivalent to an increase in delay length of 15 weeks. All estimates of β and δ are statistically significantly different from one with $p < 0.0001$. Results are nearly identical in the full sample, shown in Appendix Table A.1.

Table 2: Aggregate discount parameter estimates from quasi-hyperbolic model

Model:	OLS	Interval regression	NLS	Binary logistic
	(1)	(2)	(3)	(4)
δ	0.9980 (0.0003)	0.9979 (0.0003)	0.9981 (0.0003)	0.9983 (0.0003)
β	0.9711 (0.0032)	0.9712 (0.0033)	0.9723 (0.0031)	0.9778 (0.0031)
r (annual discount rate)	0.1098 (0.0150)	0.1146 (0.0174)	0.1066 (0.0148)	0.0901 (0.0145)
$H_0: \delta = 1$	$p < 0.0001$	$p < 0.0001$	$p < 0.0001$	$p < 0.0001$
$H_0: \beta = 1$	$p < 0.0001$	$p < 0.0001$	$p < 0.0001$	$p < 0.0001$

Notes: Standard errors are clustered at the subject level. δ is the weekly exponential discount factor, and $r = \delta^{-52} - 1$ is the annual discount rate. The data consist of 4,345 price lists from the 79 subjects in the balanced sample. The binary logistic model assumes a standard T1EV error distribution, and uses data from 95,590 binary choices.

4.2 Individual discounting estimates

When a time preference elicitation is bundled into the design of a larger research study, it is expected to deliver an individual-specific measure of discounting that is either a mediating variable or an outcome of interest. Using the OLS technique from equation 6, we successfully estimate individual-specific δ and β parameters for every subject in the sample (both balanced and full). Appendix Figure A.2 shows the distributions of estimates of r , the annual interest rate, and β , from the balanced sample.

The average (median) r estimate is 11.8% (5.7%), with 81% of subjects having $r > 0$. The average (median) β estimate is 0.972 (0.979), with 89% of subjects having $\beta < 1$. Results for the full sample are in Appendix Figure A.3. While there are some outliers in the right-tail of the discount rate distribution in the full sample, the median of 7.8% is very similar to that of the balanced sample. The median and mean β estimates are virtually identical across samples.

5 Time-variance: reduced form analysis

We now exploit the longitudinal nature of our data. Figure 4 shows how average switch points change over the course of the study, separately by price list features (delay length and front-end delay) in the balanced sample; i.e. this figure shows purely the evolution of choices from identical sets from the same subjects over time. The downward trend across survey weeks is clear in essentially all choice sets, with a very similar pattern visible in the full sample, shown in Appendix Figure A.4. This suggests that subjects are getting less patient (discounting more) over time throughout our study. This is important in our experimental context: decreasing patience is exactly the kind of time-varying time preference that would lead researchers to misclassify time-variance as present-biased time inconsistency. We return to this issue in more detail later in this section.

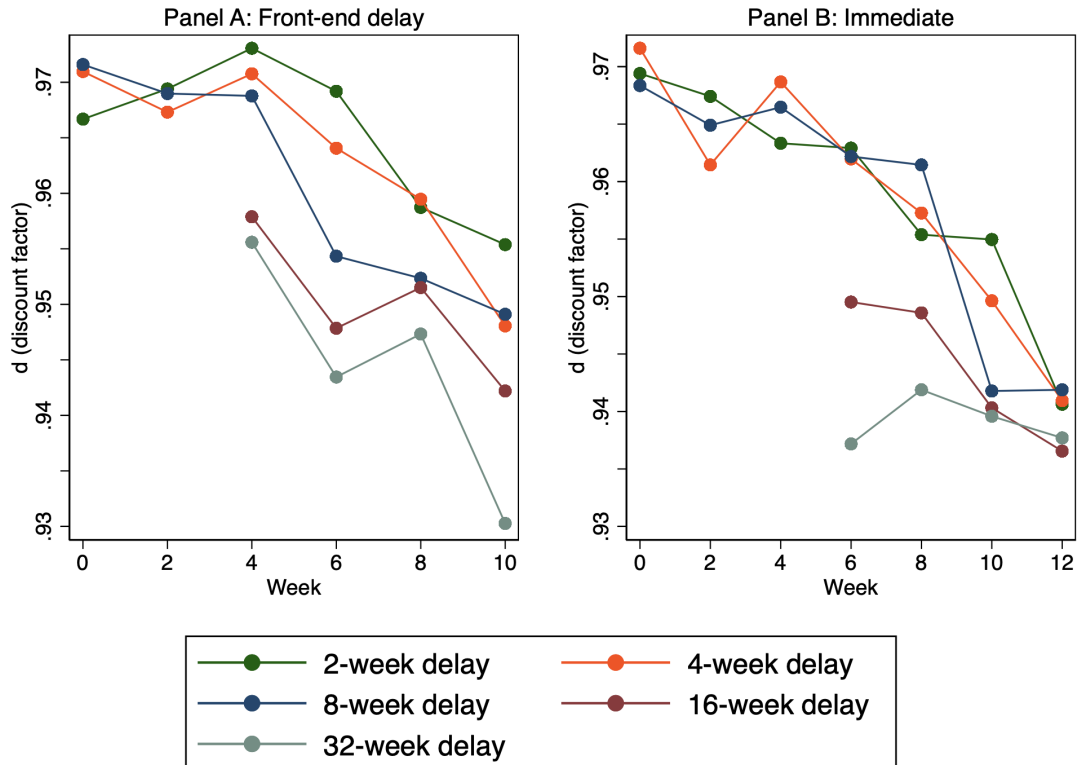


Figure 4: Time-variance in discounting

Notes: The data consist of 4,345 price lists from the 79 subjects in the balanced sample.

To assess the magnitude of these changes, we assume exponential discounting and first consider the two-week discount factor with no front-end delay in the balanced sample, as we observe this in all seven surveys of the study. The average d falls from 0.969 in Week 0 to 0.941 in Week 12. This corresponds to an increase in the annual discount rate from 124% to 391%. However, Figure 4 also showcases that discounting is less sensitive to delay length variations between two and eight weeks than the exponential model would predict, and so we consider the evolution of the annual discount rate as calculated from the eight-week factor as well. In this case, the decreasing patience over the study corresponds to an increase in the annual discount rate from 23.2% in Week 0 to 47.6% in Week 12.¹¹ In both cases, this is an economically meaningful change in discounting.

To assess the significance of these changes, we estimate OLS regressions of d on a linear survey week variable, with fully-interacted fixed effects for delay length and front-end delay, and standard errors clustered at the subject level. In the balanced sample, we estimate a decrease in the discount factor of 0.002 per week ($p = 0.001$), and in the full sample we estimate a decrease of 0.001 per week ($p = 0.017$). While the estimates are qualitatively similar, we can reject that they are the same in magnitude across samples ($p = 0.009$). This suggests that if anything, attrition may attenuate the decreasing-patience time-variance we observe in the study.

A key strength of our data collection is our ability to observe subject-level changes in discounting over time. While on average, subjects appear to be making less patient choices over time, we can characterize the distribution of time-variance. We estimate the same model as in the previous section—a linear OLS regression of the discount factor on a survey week variable with fully-interacted delay-length and front-end delay fixed effects—at the individual level. Appendix Figure A.5 shows the distributions of week coefficients in the full sample (Panel A) and balanced sample (Panel B). Negative values correspond to time-variance in the direction of decreasing patience.

Overall, 77 of 153 (50%) of subjects in the full sample, and 47 of 79 (59%) of subjects

¹¹ All of the rates are higher than those present in Table 2 because we have not yet accounted for present-bias in this section.

in the balanced sample exhibit decreasing patience over the course of the study. Among this group, we can reject time-invariance for 39 subjects (25%) in the full sample and 28 subjects (35%) in the balanced sample. 57 subjects (37%) in the full sample exhibit increasing patience, and we can reject time-invariance for 33 (22%) of them. 25 subjects (32%) in the balanced sample exhibit increasing patience, and we can reject time-invariance for 15 (19%) of them. That leaves 19 subjects (12%) in the full sample and 7 subjects (9%) in the balanced sample as exactly time-invariant.

5.1 Does attrition explain aggregate time-variance?

While subjects have a small likelihood of winning a prize each time they take a survey, the guaranteed, substantive payment for participating in the study comes after all surveys have been completed. When it comes time to participate in survey week two, that delayed reward is ten weeks in the future, and it gets two weeks closer in each successive survey. If impatient subjects attrit because they do not value the completion payment, we should expect to see a large initial decrease in discounting from survey weeks zero to two (because week zero is incentivized on its own), and then a stable pattern thereafter.¹² This is not the pattern of time-variance we find in the full sample shown in Appendix Figure A.4. Furthermore, Appendix Figure A.6 shows how the discount factor within each survey correlates with whether a subject does not complete the following survey. Specifically, for each survey we regress the discount factor on an indicator variable for whether a subject fails to complete the next survey, with fully-interacted fixed effects for delay length and front-end delay, and standard errors clustered at the subject level. Plotting the attrition coefficients and their 95% confidence intervals shows that none of the estimates are statistically significant, and the sign is not consistent across surveys.

¹²Purely from a patience perspective, if week two is worth completing for the reward in ten weeks, then week four is worth it for the reward in eight weeks.

5.2 Time inconsistency and non-stationarity

In Section 2, we illustrated how time-varying behavior implies either non-stationary or time inconsistent behavior (or both). We have now empirically established both decreasing patience and inconsistent/non-stationary behavior in the data. However, we have not yet drawn a distinction between time inconsistency and non-stationarity. In this section, we allow them to differ and explore the connection between them and with time-variance; how much observed inconsistency is explained by time-variance rather than non-stationarity?

We start by separately estimating non-stationarity and inconsistency. Recall that in Table 1, the “Immediate” coefficient represented the effect of removing a front-end delay on the discount factor. By adding *survey-week* fixed effects to these models we isolate only within-survey variation in the existence of a front-end delay, making the Immediate coefficient a measure of non-stationarity alone. By instead adding *drawing-week* fixed effects, we isolate only across-survey variation in the existence of a front-end delay, making it a measure of inconsistency alone. Specifically, the drawing week of a decision is the week the *earlier* drawing takes place. For example, decisions in the first survey with a two-week front-end delay have the same drawing week as decisions in the second survey without the front-end delay. Estimates are in Table 3 for the balanced sample, and in Appendix Table A.2 for the full sample.

The pattern of results is striking: our observation of present-bias in the data is entirely due to violations of time-consistency rather than stationarity. We find no statistically significant relationship between a *within-survey* front-end delay and the discount factor (e.g. $p = 0.347$ for the linear model in the balanced sample, $p = 0.537$ for the linear model in the full sample). On the other hand, we find that making the same choice two weeks in advance (*within-drawing-week*) results in a significantly higher discount factor (e.g. $p < 0.001$ for the linear model in the balanced sample, $p = 0.004$ for the linear model in the full sample). These findings are consistent with Section 2.1, where apparent time-inconsistent present bias is a result of decreasing patience under the assumption of stationarity.

We follow Janssens et al. (2017) in asking how much inconsistency and non-stationarity

Table 3: Separate estimates of non-stationarity and inconsistency

Measure:	Non-stationarity		Inconsistency	
Dependent variable:	d	$\ln(d)$	d	$\ln(d)$
	(1)	(2)	(3)	(4)
Immediate	-0.001 (0.001)	-0.001 (0.002)	-0.006*** (0.002)	-0.007*** (0.002)
Constant (Delay = 8 weeks, Immediate = 0, Week = 0)	0.969 (0.004)	-0.033 (0.005)	0.975 (0.005)	-0.026 (0.005)
Survey-week FE	Y	Y	N	N
Drawing-week FE	N	N	Y	Y
Clusters	79	79	79	79
Observations	4,345	4,345	4,345	4,345

Notes: OLS regressions with standard errors clustered at the subject level. All models include delay-length fixed effects with eight weeks as the omitted group. Week zero is always is omitted group from the survey-week and drawing-week fixed effects. Data are limited to the balanced sample.

in our study can be attributed to time-variance. Using the decision triangle language of Section 2.1, given any time invariance ($a - c$), we can calculate whether observed inconsistency ($b - c$) is exactly predicted by, in excess of, or less than what would be predicted assuming stationarity holds ($a = b$). Specifically, for all decision triangles with inconsistency ($b \neq c$), we calculate $(a - c)/(b - c)$. This is effectively the percentage of time inconsistency that is attributable to time-variance. *Time-variance explains 72% of the inconsistency in our sample.* While inconsistency is typically attributed to a behavioral present-bias, this result suggests that in a typical experimental context a large part of it could be explained by decreasing patience instead. We can do the same for any observed non-stationarity as well. For all decision triangles where non-stationarity is present ($a \neq b$), we construct a similar ratio of $(a - c)/(a - b)$. This ratio describes the percentage of non-stationarity that can be attributed to time-variance. Time-variance explains 56% of non-stationarity in our sample, however non-stationarity is less common in our data.

5.3 Time preference classification

We now follow Halevy (2015) and classify each subject into one of the five possible combinations of time preference properties. We start by classifying each decision within a triangle as either time-invariant or time-varying. This requires that subjects must have attended at least two sequential surveys to be included in this analysis. Out of 153 total subjects, this applies to 141¹³. These 141 subjects account for 6,267 of the 6,367 price lists collected in our study, containing 2,506 decision triangles. Of these 141 subjects, 130 subjects (92%) make a time-varying decision at some point in the study. Of the 2,506 decision triangles, 1,500 (60%) include a time-varying decision¹⁴. Both the percentage of subjects and percentage of choices that are time-varying exceed the findings of Halevy (2015), that found a maximum of 56.42% of subjects exhibiting time-varying preferences. While the longer longitudinal dimension of our study contributes to that difference at the subject level, this should not be the case at the triangle level. We return to this issue later in this section.

We count the occurrences of inconsistent and non-stationary decisions in a similar manner to the time-varying decisions. While the same 141 subjects and 6,267 price lists generate 2,710 decision triangles where b and c the test for inconsistency is possible and 2,782 decision triangles where the test for non-stationarity is possible, we restrict our analysis to the 2,506 decision triangles that overlap with the preceding time-invariance analysis. We find that 129 subjects (91%) make an inconsistent decision at some point, with 1,448 (58%) of triangles being inconsistent. Similarly, 127 subjects (90%) make a non-stationary decision at some point, with 1,276 (51%) triangles in total being non-stationary. Both proportions exceed those from Halevy (2015) again, where at most 66.33% of subjects exhibited inconsistent preferences and at most 55.22% exhibited non-stationary preferences.

Table 4 shows the resulting classification of subjects by property in our sample in columns (1) and (3) of Panel A for the full and balanced samples, respectively. Focus-

¹³This eliminates ten subjects who only responded to the first survey and two more subjects who only responded to the first and third.

¹⁴This proportion is very similar if we also consider time-variance in the choices made *with* a front-end delay. In this case, 94% of subjects display time-varying behavior in 59% of triangles.

ing on the balanced sample, only eight subjects (10%) exhibit invariant preferences. Of these eight, seven (9% of the sample) make consistent and stationary decisions as well. However, these seven subjects display constant discounting, not reacting to differences in choice delay or front-end delay either. Thus, among subjects who reacted to any stimuli, all violated a time preference property. 69 (87%) subjects violate each property at some point, two (3%) maintain only stationarity, and none maintain only consistency.

Table 4: Subject- and decision-triangle-level classification of time preference properties

Sample:	Full		Balanced	
Margin of Error:	None	± 6 tickets	None	± 6 tickets
Properties of choices	(1)	(2)	(3)	(4)
<i>Panel A: Subject level</i>				
A) Invariant, stationary, consistent	6.38%	20.57%	8.86%	20.25%
B) Invariant only	1.42%	0.00%	1.27%	0.00%
A) + B)	7.80%	20.57%	10.13%	20.25%
C) Stationary only	3.55%	7.09%	2.53%	3.80%
D) Consistent only	2.13%	1.42%	0.00%	0.00%
E) None	86.52%	70.92%	87.34%	75.95%
C) + D) + E)	92.20%	79.43%	89.87%	79.75%
<i>Panel B: Triangle level</i>				
A) Invariant, stationary, consistent	33.88%	61.61%	37.45%	63.40%
B) Invariant only	6.26%	7.30%	6.07%	7.01%
A) + B)	40.14%	68.91%	43.52%	70.41%
C) Stationary only	15.20%	15.20%	15.24%	14.40%
D) Consistent only	8.34%	8.30%	7.86%	7.96%
E) None	36.31%	7.58%	33.39%	7.23%
C) + D) + E)	59.86%	31.09%	56.48%	29.59%

Notes: Columns (1) and (2) in Panel A use the 141 subjects that completed consecutive surveys. Columns (3) and (4) use the 79 subjects that completed all surveys. Columns (1) and (2) in Panel B use all valid decision triangles in the study, and columns (3) and (4) in Panel B use all decision triangles in the balanced sample. Columns (1) and (3) assume that any deviation in choices across price lists with equivalent delay lengths is a time preference property violation. Columns (2) and (4) assume that any deviation greater than six tickets from an analogous choice is a time preference violation, capturing a range of a standard deviation around a choice ($\sigma/2 \approx 7$, which due to interval of two tickets between choice rows corresponds to equality within six tickets).

A key caveat to these statistics is that the prevalence of property violations in our sample can be, in part, attributed to the large number of decisions we offer subjects. Our study design allows for formal modeling and testing of each property at the individual level, which we do in the next section. However, we first address this issue with two ad-hoc approaches that allow us to continue to compare our results to other studies. First we allow for a margin of error that spans a standard deviation (14 tickets) of the decisions we observe (± 6 tickets due to the two-ticket gap between each choice row). If a subject's choice is within six tickets on either side of their previous choice, we treat the decision as if it did not violate the tested time preference property.¹⁵ Results are in columns (2) and (4) of Panel A in Table 4. Second, instead of aggregating to the subject-level, we treat each decision triangle as a distinct observation. Results are in columns (1) and (3) of Panel B, and results using both modifications are in columns (2) and (4) of Panel B.

Using the margin of error at the subject level, 16 subjects (20%), exhibit time-invariant behavior. All 16 of these subjects are also consistent and stationary. The increase in the size of this group comes mostly from a reduction in size of those that violate every property, now 60 subjects (76%). Three subjects maintain only stationarity (4%) and still none maintain only consistency using this approach. Taking the decision-triangle level of analysis approach instead we find that around 43% of decision triangles maintain time-invariance. This is close to the minimum classification from Halevy (2015), which has 44% of subjects exhibiting time-invariance. Our most conservative estimates of time-varying behavior come from combining the decision triangle sorting with the margin of error. This method shows 70% of decision triangles maintain time-invariance, which is close to the maximum classification from Halevy (2015), which has 68% of subjects exhibiting time-invariance. Overall, these results suggest that observing a single decision triangle per individual understates the potential for time-varying behavior. Defined strictly, time-variance is a universal feature of subjects who responded to our stimuli.

¹⁵One problem with adding a margin of error to this accounting exercise is that it is possible for only one property to be violated, even though theoretically, when two hold, the third is guaranteed. When this is the case, we classify a decision triangle or subject as satisfying all time preference properties.

6 Building a time-varying discount function

To account for time-varying preferences the limited existing literature has used reduced-form approaches to correcting $b - c$ in light of $a \neq c$, or (less-likely given design constraints) correcting $a - b$ in light of $a \neq c$ (e.g. Janssens et al. (2017) or Harrison et al. (2025)). We offer an alternative: a *potentially* time-varying discount function. This section maps the five groups of time preference properties to structural discount functions and their properties.

Time-invariant preferences are well represented in the literature. Discount functions that admit these preferences include the standard exponential, as well as the hyperbolic and β - δ models regularly used to model time-inconsistent decisions. These models all share a common property: They are a function of only the distance between payoff time and evaluation time, $t - \tau$. Thus, $D(\tau, t)$ is typically reduced to just $D(t)$, where τ is always 0, without loss of generality. This characterizes the necessary and sufficient condition for a function to be time-invariant.

Proposition II. *A discount function admits time-invariant preferences if and only if $D(\tau, t) = D(t - \tau)$.*

See Appendix Section A.2 for the proof.

We are only aware of one discounting functional form in the literature that violates this property. Strulik (2021) introduced a “time-consistent hyperbolic” discount function which violates the assumption of time-invariance:

$$D(\tau, t) = \left(\frac{1 + \alpha\tau}{1 + \alpha t} \right)^\beta \quad \alpha > 0, \beta > 1. \quad (7)$$

Strulik (2021) designed the function for use in environmental resource models, relying on the increasing patience mandated by the model to avoid extinction steady states. The ability to commit to consumption plans in his context requires that the discount function maintains time-consistency. Therefore, this model must be only time-consistent, as it is not time-invariant and so cannot be both time-consistent and stationary. In general, what makes

a discount function time-consistent? Strotz (1956) suggests that the only discount function that can achieve time-consistency is the standard exponential. However, this result relies on the assumption of a discount function that is a function of only $t - \tau$. Strotz's result has since been generalized to prove instead that time-consistency of a discount function is equivalent to multiplicative separability in t and τ (Burness, 1976; Drouhin, 2020).

Proposition III. *A discount function $D(\tau, t)$ admits consistent preferences if and only if it can be written:*

$$D(\tau, t) = f(t) \cdot g(\tau)$$

Further, if the discount function can be written in this way, then $g(x) \equiv \frac{1}{f(x)}$.

Including the time-consistent hyperbolic model, three of the five possible time preference groups have been characterized by discount functions in the literature. To our knowledge, discount functions have not been developed to fit the two remaining groups: only stationarity and adhering to none of the TICS properties. To develop these functions in the next section, we present the necessary and sufficient conditions for a discount function to admit stationarity, which also, to our knowledge, has not been shown in the literature.

Proposition IV. *Let $\alpha \in \mathbb{R}^+$. A discount function admits stationary preferences if and only if it takes the form*

$$D(\tau, t) = \alpha^{(t-\tau)f(\tau)}.$$

See Appendix Section A.3 for the proof.¹⁶

With propositions II, III, and IV, we arrive at the following well-known result:

Corollary. *The discount function $D(\tau, t)$ admits preferences exhibiting stationarity, time-consistency, and time invariance if and only if it is the exponential discount function.*

That is, $D(\tau, t)$ can be written as this function of $t - \tau$ alone: $D(t - \tau) = \delta^{\beta(t-\tau)}$.

¹⁶Note that f may be trivial in τ .

6.1 Nested Exponential Discount Function

With necessary and sufficient conditions for a discount function to satisfy any of the TICS properties, we now have the tools to develop a discount function that can fit all five possible classes of time preferences. Two goals motivate our approach to developing such a function. First, our discount function should be fully general in its admittance of time preference properties. For instance, the β - δ model acts as an extension of the standard exponential, allowing for time-inconsistent and non-stationary preferences by setting $\beta \neq 1$. By extending the standard exponential in a similar manner, we can achieve each of the five time preference classes according to its parameter values. Second, a time-varying discount function should be capable of exhibiting both increasing or decreasing patience depending on parameter values. The time-consistent hyperbolic model from Strulik (2021) only allows for increasing patience, which was appropriate contextually. However, allowing for decreasing patience is important for making the function generally applicable.

We introduce the following discount function:

$$D(\tau, t) = \delta^{\frac{(t^\mu - \tau^\mu)^\eta}{\gamma^\tau}} \quad \eta > 0, \gamma > 0, \mu > 0 \quad . \quad (8)$$

This function, which we call the *nested exponential discount (NED) function*, can satisfy any of the five possible arrangements of the TICS properties. Suppose $\eta = \gamma = \mu = 1$. Then the NED is identical to the standard exponential discount function. Now suppose $\eta = \gamma = 1$ while $\mu \neq 1$. Then the function is no longer a function of only $t - \tau$ nor does its exponent satisfy the conditions required to maintain stationarity. However, the function does remain multiplicatively separable in t and τ , granting it time-consistency only. The argument follows similarly for when $\eta = \mu = 1$ and $\gamma \neq 1$. In this case, the function is not a function of only $t - \tau$ and is no longer multiplicatively separable in t and τ . Its exponent, however, can be expressed as $(t - \tau)f(\tau)$, granting stationarity only. When $\gamma = \mu = 1$, but $\eta \neq 1$, NED is a discrete-time reformulation of the Weibull discount function, a less common behavioral model that achieves the same violations of the TICS properties as

the hyperbolic and β - δ models. Specifically, in this case we have that the function is not multiplicatively separable in t and τ , the exponent is not of the form $(t - \tau)f(\tau)$, but the function only depends on $t - \tau$. Since any of these three parameters not equaling one results in the violation of two properties, whenever two or more parameters are not equal to one, none of the properties are satisfied. Therefore, all five groups of time preference properties can be modeled with this one function. Note that the function can also be expressed as

$$D(\tau, t) = \exp\left(-\rho \cdot \frac{(t^\mu - \tau^\mu)^\eta}{\gamma^\tau}\right), \text{ where } \rho = -\ln(\delta) \quad , \quad (9)$$

similar to the alternate format of exponential discounting.

The NED function allows for increasing or decreasing patience depending on the parameter values of γ and μ . For instance, when $\mu = 1$, values of $\gamma < 1$ correspond to decreasing patience. As τ increases, the relative distance between t and τ is amplified by a denominator that is increasingly less than one, in turn increasing the exponent, and therefore decreasing the discount factor. The opposite holds for values of $\gamma > 1$. Similarly, when $\gamma = 1$, values of $\mu < 1$ correspond to increasing patience. Holding fixed the distance $t - \tau = C$, the term $t^\mu - \tau^\mu$ is decreasing in τ , resulting in a discount factor increasing in τ . The opposite again holds for $\mu > 1$. Since η on its own cannot cause increasing or decreasing patience, it only acts to mitigate or amplify the effects of the other parameters when active at the same time. The exact condition that dictates the direction of changing patience over time of the nested exponential discount function is given by:

$$D(\tau, \tau + \Delta) > D(\tau', \tau' + \Delta) \iff \frac{\ln(\gamma)}{\eta\mu} > \frac{(\tau + C)^{\mu-1} - (\tau)^{\mu-1}}{(\tau + C)^\mu - (\tau)^\mu} \quad . \quad (10)$$

The right side is positive (negative) for $\mu > 1$ ($\mu < 1$), while the left is positive (negative) for $\gamma > 1$ ($\gamma < 1$). Therefore, the NED function predicts strictly increasing patience, $\frac{\partial D}{\partial \tau}|_{t-\tau=C} > 0$, when $\gamma > 1$ and $\mu < 1$, and $\frac{\partial D}{\partial \tau}|_{t-\tau=C} < 0$ when $\gamma < 1$ and $\mu > 1$.

Figures 5, 6, and Appendix Figure A.7 visualize the roles of each parameter in the NED function. The time-invariant behavior of the function is shown in Figure 5, which

highlights that it is effectively another approach to hyperbolic-style discounting, with the η parameter governing the direction and degree of distortion from exponential discounting.

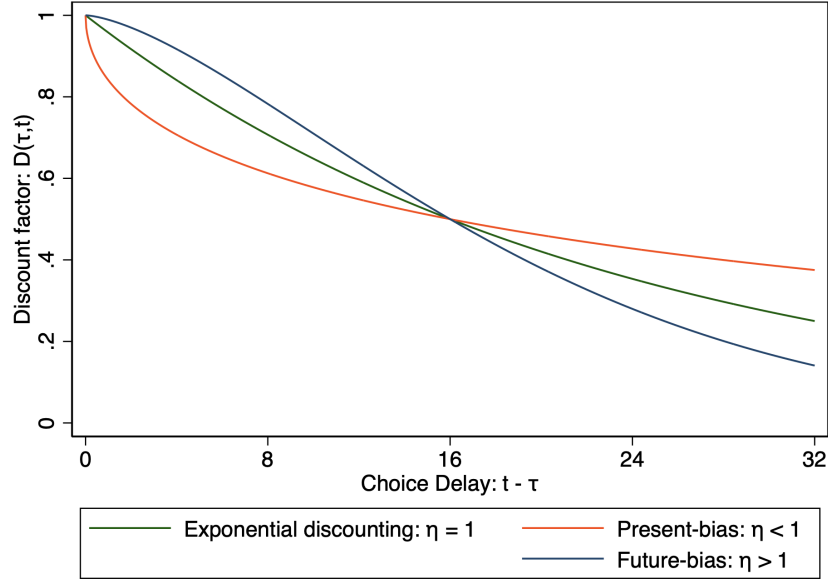


Figure 5: Time-invariant NED function ($\mu = \gamma = 1$)

Notes: the value of δ in each function is chosen such that $D(\tau, \tau + 16) = 0.5$. We show $\eta = 0.5$ and $\eta = 1.5$.

The time-consistent behavior of the function is shown in Appendix Figure A.7, but it can be simply described in relation to the time-invariant effects of η . The μ parameter can create similar non-stationary present/future biases as η , but as time passes, those biases will erode. Panel A shows that μ and η have identical effects when $\tau = 0$, but that as τ increases to two in Panel B and four in Panel C, the distortion away from exponential discounting caused by $\mu \neq 0$ becomes smaller. The intuition here is that for small τ relative to t , the effect of μ is similar to η , but as τ approaches t , this effect diminishes. For example, an agent that is “present-biased” according to a stationarity violation at $\tau = 0$ will get more patient over time as they maintain time-consistency.

The time-stationary ($\eta, \mu = 1$) behavior of the NED function is shown in Panel A of Figure 6. We set δ so that for each function $D(0, 8) = 0.5$, and then show the effects of τ increasing as time passes. γ determines whether an agent becomes more or less patient

over time, and thus whether individuals would appear “present-biased” or “future-biased” in a violation of time-consistency. Panel B shows that when the function obeys none of the three properties, non-monotonic time-variance can emerge through the interaction of μ and γ , but that γ ultimately controls the long-run direction of preference drift.

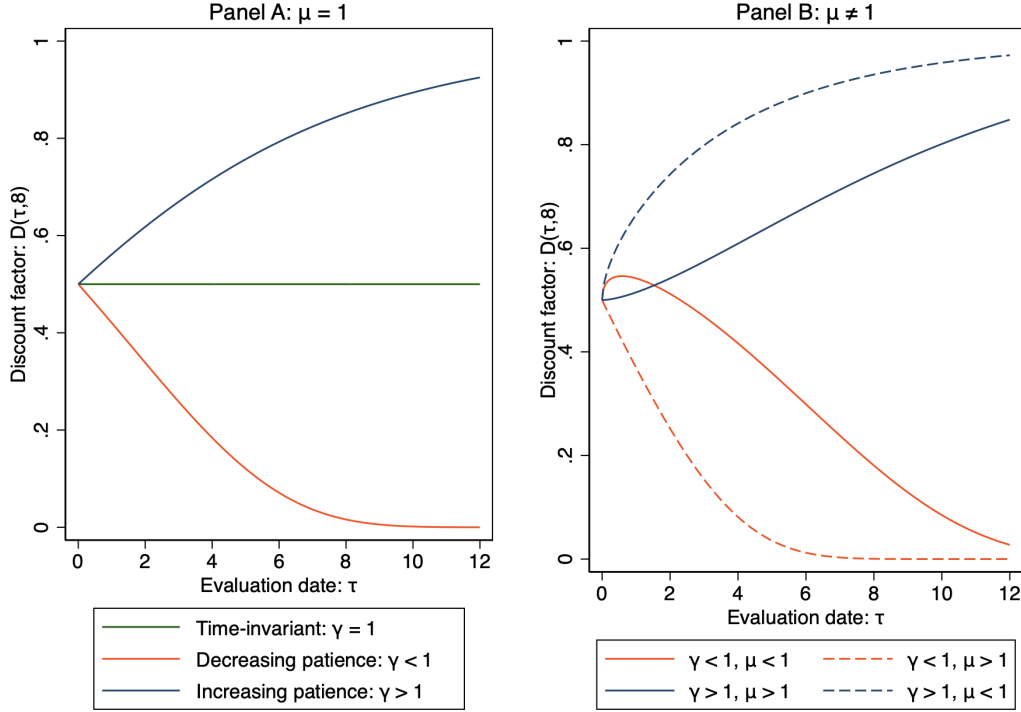


Figure 6: Time-variance in the NED function for an eight-week choice delay

Notes: the value of δ in each function is chosen such that $D(0, 8) = 0.5$. We show $\gamma = 0.8$ and $\gamma = 1.2$ in both panels, and $\mu = 0.5$ and $\mu = 1.5$ in Panel B. $\eta = 1$ in both Panels.

This non-monotonicity is an extreme example of the fact that when it comes to estimating time-invariance and the parameters of nested exponential discount function, the longitudinal dimension of the data determines the precision and reliability of those estimates. Existing work by Halevy (2015) and Janssens et al. (2017) collects data on a single decision triangle (ignoring different prices or choice delays within a triangle).¹⁷ We instead design

¹⁷Harrison et al. (2025) collects longitudinal data without overlapping dates and uses structural assumptions to analyze them like a triangle.

a study to observe repeated observations of triangles for every subject. Time-variance need not be idiosyncratic subject-level noise at different dates; instead it can be systematic learning (about either the decisions themselves or preferences over them), changes to the macroeconomic environment, the psychological environment, or preference evolution.

7 Time-variance: structural analysis

We first estimate discount function parameters at the aggregate sample level. We use the non-linear least squares method, and estimate both restricted and unrestricted versions of the NED model. Results and descriptive statistics of these estimations on the balanced sample are presented in Table 5. We limit the sample to the set of choices that were available in all weeks of the study to maintain a clear interpretation of time-variance in the absence of a flexible set of fixed effects. Column (1) enforces exponential discounting ($\mu = \eta = \gamma = 1$), column (2) enforces time-invariance only ($\mu = \gamma = 1$), column (3) enforces stationarity only ($\mu = \eta = 1$), column (4) enforces consistency only ($\eta = \gamma = 1$) and column (5) makes no parameter restrictions. Estimates using the full sample are in Appendix Table A.3.

Besides direct tests of the estimated parameters for adherence to the time preference properties, we assess the fit of each version of the model. We use both the Akaike Information Criteria (AIC), which applies a parameter penalty to model likelihoods, and Bayesian Model Averaging (BMA) to do this.¹⁸ For each estimated model we report the AIC, model selection weights calculated using the AIC (Burnham and Anderson, 2004), and the posterior model probabilities (PMPs) from the BMA procedure (Hoeting et al., 1999).¹⁹ Both the AIC weights and Bayesian PMPs are measures of relative model fit (from 0 to 1) among the set of five models in the table.

¹⁸The AIC is defined as $2k - 2\ln(\mathcal{L})$, where k is the number of model parameters and $\mathcal{L}$ is its fitted likelihood, meaning that lower AICs represent a better fit. BMA instead assigns posterior probabilities to candidate models according to how well they fit the data (relative to their priors, which we assume are equal), and combines the results across models.

¹⁹The AIC model selection weights are calculated with respect to the set of models estimated as $w_i = \frac{\exp(-\Delta_i/2)}{\sum_j \exp(-\Delta_j/2)}$, for all models j , and $\Delta_i = AIC_i - \min_j(AIC_j)$. The Bayesian PMPs are the posterior probability from the BMA exercise.

Table 5: Aggregate parameter estimates for the nested exponential model

Restrictions:	$\mu = \eta = \gamma = 1$	$\mu = \gamma = 1$	$\mu = \eta = 1$	$\eta = \gamma = 1$	None
Properties					
Invariant	✓	✓	X	X	X
Stationary	✓	X	✓	X	X
Consistent	✓	X	X	✓	X
	(1)	(2)	(3)	(4)	(5)
δ	0.9932 (0.0007)	0.9807 (0.0023)	0.9953 (0.0007)	0.9958 (0.0017)	0.9843 (0.0027)
η	1 .	0.5394 (0.0309)	1 .	1 .	0.5838 (0.0547)
γ	1 .	1 .	0.9401 (0.0152)	1 .	0.9401 (0.0213)
μ	1 .	1 .	1 .	1.1524 (0.1255)	0.8619 (0.1509)
AIC	25,072	24,959	25,017	25,068	24,914
AIC weight	< 0.0001	< 0.0001	< 0.0001	< 0.0001	1.000
Bayesian PMP	< 0.0001	< 0.0001	< 0.0001	< 0.0001	1.000
r_0	0.4265 (0.0524)	0.1786 (0.0230)	0.2785 (0.0482)	0.4976 (0.0913)	0.1223 (0.0243)
r_{12}	$= r_0$	$= r_0$	0.6745 (0.1131)	0.5504 (0.1363)	0.2583 (0.0450)
PB_S	1 .	0.9795 (0.0026)	1 .	1.0043 (0.0029)	0.9997 (0.0129)
PB_C	1 .	$= PB_S$	0.9950 (0.0010)	1 .	0.9800 (0.0023)
$H_0: \delta = 1$	$p < 0.0001$	$p < 0.0001$	$p < 0.0001$	$p = 0.0142$	$p < 0.0001$
$H_0: \eta = 1$	.	$p < 0.0001$	.	.	$p < 0.0001$
$H_0: \gamma = 1$	.	.	$p = 0.0002$	.	$p = 0.0062$
$H_0: \mu = 1$	.	.	.	$p = 0.2283$	$p = 0.3630$
$H_0: \mu = \gamma = 1$	.	.	.	.	$p = 0.0055$
$H_0: \mu = \eta = 1$	.	.	.	.	$p < 0.0001$
$H_0: \eta = \gamma = 1$	.	.	.	.	$p < 0.0001$
$H_0: \mu = \eta = \gamma = 1$	.	.	.	.	$p < 0.0001$
$H_0: r_0 = r_{12}$	.	.	$p = 0.0009$	$p = 0.2991$	$p = 0.0044$
$H_0: PB_S = 1$	.	$p < 0.0001$	.	$p = 0.1360$	$p = 0.9836$
$H_0: PB_C = 1$	.	$p < 0.0001$	$p < 0.0001$	.	$p < 0.0001$
$H_0: PB_S = PB_C$	.	.	.	.	$p = 0.1329$

Notes: Standard errors are clustered at the subject level. $r_\tau = D(\tau, \tau + 52)^{-1} - 1$ is a measure of the annual discount rate. PB_S and PB_C are stationarity-based and consistency-based measures of present-bias. The data consist of 3,081 price lists from 79 subjects that completed all seven surveys, using only the choice sets that spanned every survey. All estimates are from non-linear least squares regressions.

In column (1) the NED model is fully restricted to exponential discounting, and we obtain an estimate of the weekly discount factor of 0.9932. This translates into an annual discount rate (r_0) of 43%. The model in column (2) allows for time-inconsistency and non-stationarity, but enforces time-invariance, similar in spirit to quasi-hyperbolic discounting, but with a functional form more closely related to hyperbolic discounting. The estimate of $\delta = 0.9807$ suggests much more discounting of near-term future outcomes than the exponential estimate in column (1), but the “present-bias” parameter η flattens the discount function for longer choice delays. $\eta = 0.5394$ turns a delay length of two weeks into an effective delay of 1.5 weeks, and a 52-week delay length into an effective delay of 8.4 weeks, such that the annual discount rate for a one-year choice delay is estimated to be much lower than in the exponential model: 18%. By comparison, without the hyperbolic “flattening” of the discount factor in this function, a weekly discount factor of 0.9807 would exponentially imply an annual discount rate of 175%. This discrepancy means we estimate significant present bias in discounting, clearly rejecting $\eta = 1$ ($p < 0.0001$), and that this model offers a substantial improvement in fit (lower AIC) when accounting for it. To assess the magnitude of present-bias estimated by the model in a familiar metric, we construct measures PB_S and PB_C , which are discount factor ratios that correspond to how researchers would typically measure a quasi-hyperbolic β through either a stationarity violation or a consistency violation, respectively. Specifically,

$$PB_S = \frac{D(0, 8)}{D(0, 0)} / \frac{D(0, 10)}{D(0, 2)}, \text{ and } PB_C = \frac{D(2, 10)}{D(2, 2)} / \frac{D(0, 10)}{D(0, 2)}. \quad (11)$$

When invariance holds these measures are identical, and for the model in column (2) we estimate $PB_S = PB_C = 0.9795$, which is significantly different from one ($p < 0.0001$).

The model in column (3) allows for time-variance and inconsistency, but enforces time-stationarity, the property specification that does not appear in the literature. The estimate of the weekly discount factor when $\tau = 0$ is 0.9953, which translates to an annual rate (r_0) of 28%. Time-variance due to $\gamma = 0.9401$ ($p = 0.0002$ tested against one) decreases that weekly discount factor as the study progresses –the annual discount rate at the end of

our study (r_{12}) grows to 67%, meaning that this model estimates statistically significant, decreasing-patience time-variance ($p = 0.0009$ for the test of $r_0 = r_{12}$). A direct implication of this is that the model predicts statistically significant, consistency-violation present-bias ($p < 0.0001$ for the test of $PB_C = 1$), although the magnitude is very small. While this model captures the time-variance present in our data, it does not feature hyperbolicity in discounting; in net this tradeoff means it fits better than the exponential model, but not as well as the invariant-only model in column (2).

The model in column (4) allows for time-variance and non-stationarity, but enforces time-consistency, the “time-consistent hyperbolic” property pattern studied in Strulik (2021), but in a more flexible form. The initial ($\tau = 0$) weekly discount factor estimate is 0.9958, with the hyperbolicity parameter $\mu > 1$ steepening the discount function for longer choice delays. $\mu = 1.1524$ is not significantly different from one ($p = 0.2283$), and turns the choice between zero and two weeks into an effective delay of 2.2 weeks, and the choice delay between zero and 52-weeks into an effective delay of 95.0 weeks, yielding an initial ($\tau = 0$) estimate of the annual discount rate of 50%. The end-of-study ($\tau = 12$) annual discount rate predicted by this model is slightly higher at 55%, which we cannot reject is equal to the initial rate ($p = 0.2991$). The μ parameter acts on τ as well as t , thus while the effective choice delay between zero and 52 weeks is 95.0 weeks, the effective choice delay between 12 and 64 weeks is 103.1 weeks. A direct implication of decreasing patience in this model should be stationarity-violation future-bias, which we estimate, albeit to an insignificant degree ($p = 0.1360$). The μ parameter in this model is forced to “choose” between fitting two features of the data that, according to the model, cannot co-occur, and thus it fits poorly. We fail to reject it is different than the exponential discounting model, and the model offers only a marginal improvement in fit relative to column (1).

The model in column (5) puts no restrictions on the parameters. We estimate an initial weekly discount factor of 0.9843, and an initial annual discount rate of 12% (this rate depends on δ , η , and μ). This grows to 26% over the course of the study, reflecting the decreasing patience we identified in the reduced-form analysis. We can reject a time-invariant an-

nual discount rate ($p = 0.0044$). Using the NED function parameters to test for adherence to the TICS properties, we can reject that all three properties hold ($p < 0.0001$), that time-invariance holds ($p = 0.0055$), that stationarity holds ($p < 0.0001$), and that consistency holds ($p < 0.0001$). On its own, we fail to reject $\mu = 1$ ($p = 0.3630$), which suggests we could adopt a more parsimonious version of the function where no properties hold. We reject that both $\gamma = 1$ ($p = 0.0062$) and $\eta = 1$ ($p < 0.0001$). Consistent with the reduced-form results, this model predicts more consistency-violation present-bias ($PB_C = 0.9800 < 1$, $p < 0.0001$) than stationarity-violation present-bias ($PB_S = 0.9997 < 1$, $p = 0.9836$), although we marginally fail to reject that they are the same ($p = 0.1329$). Despite the AIC measure's penalty for extra parameters, the improvement in fit for the unrestricted model is such that it gets 100% of the AIC weight among the models we estimate in Table 5. The BMA procedure leads to the same result. When we restrict the model to $\mu = 1$ we do achieve a slight improvement in parameter-adjusted fit, with the restricted model assigned a weight of 0.67 using the AIC technique in a two-model comparison with the unrestricted model in column (5). Results in the full sample, shown in Appendix Table A.3 are essentially the same, with the one notable difference being that we can reject equivalent consistency-violation and stationarity-violation present-bias in the full sample.

While the focus of the current paper is on time-variance, we want to also highlight the key role of hyperbolicity in explaining our model fitting results. We also compare the invariant-only, two-parameter restricted version of the NED model in column (2) to the workhorse invariant only, two-parameter quasi-hyperbolic model. We find that the quasi-hyperbolic model does not fit the data as well, achieving an AIC of 24,984 in the sample corresponding to the estimates in Table 5, which gives it an AIC weight of zero in a two-model comparison with the model in column (2). This is because hyperbolicity is a real feature of our data. The single parameter hyperbolic discount function $D(\tau, t) = \frac{1}{1+\alpha \cdot (t-\tau)}$ achieves an AIC of 25,066 in this sample, which gives it a AIC weight of 0.97 in a two-model comparison with the exponential discounting model in column (1).

7.1 Subject-level estimates

The literature on time-preference estimation and the importance of present-bias places a strong emphasis on allowing for heterogeneity in preferences. For example, if one-in-five people are present-biased discounters we may not estimate a representative quasi-hyperbolic discount function with β significantly less than one, but we still may want to identify and target the sub-population for whom $\beta_i < 1$. In this vein, we use our non-linear least squares technique to estimate individual-specific parameters of the NED function.

We plot the distribution of parameter estimates for the balanced sample in Appendix Figure A.8, limiting the sample to only choices subjects made in every week of the study to maintain comparability with the aggregate estimates. The model converges for 59 of 79 subjects. The median subject in the balanced sample has a weekly δ of 0.9837, corresponding to an (initial: $\tau = 0$) annual discount rate of 21% (recall this rate depends on δ , η , and μ). Only one of 59 (2%) subjects is classified as a negative discounter. “Present-bias” as measured by the η parameter is common, with a median η in this sample of 0.4876, and 48 of 59 (81%) of subjects having $\eta < 1$. Estimates of γ and μ are both centered around one, with the median subject showing monotonically decreasing patience ($\gamma < 1$ and $\mu > 1$).

To make across-model fit comparisons when each model is estimated on small individual datasets ($N = 55$ at most), we use only the balanced sample, use all available choices, and implement an extra penalty for parameters in small-sample AIC calculations (Burnham and Anderson, 2002). Using the AIC we determine which model best fits each individual in sample, with the results in column (1) of Table 6. We find non-trivial fractions of the sample are best fit by the unrestricted NED function (38%), the invariant-only restricted model (35%), and the stationary-only restricted model (18%). This classification does not take into account the degree of advantage the best-fit model has over the next-best option for any given subject, so we also present tabulations of models that are classified with AIC weight > 0.5 and 0.9 in columns (2) and (3), respectively. There is no guarantee we identify a model that exceeds either of these thresholds, and the fraction of subjects that we cannot classify this way is shown at the bottom of the table. 87% of individuals yield a model

with weight above 0.5, but only 29% yield a model with weight above 0.9. The results in column (2) are similar to column (1): the unrestricted model (35%), invariant only model (33%) and stationary only model (17%) are the dominant classifications. The exponential model is never assigned an AIC weight above 0.5. In column (3) we see that while over two-thirds of the sample does not have a model that dominates the classification with an AIC weight above 0.9, most of those who can be classified are sorted into the unrestricted NED model (18%). The consistent only model is never assigned an AIC weight above 0.9. To summarize, we see that three property arrangements can effectively characterize our sample: “present bias” in which stationarity and consistency violations coincide while time-invariance holds, stationary only discounting which has no precedent in the literature to our knowledge, and discounting which obeys none of the three TICS properties.

Table 6: Subject-level structural classification within the nested exponential model

Metric:	Best fit	AIC weight > 0.5	AIC weight > 0.9
Model restrictions	(1)	(2)	(3)
Exponential	2.78%	0%	0%
Invariant only	34.72%	33.33%	1.39%
Stationary only	18.06%	16.67%	9.72%
Consistent only	6.94%	2.78%	0%
Unrestricted	37.50%	34.72%	18.06%
Not classified given metric	0%	12.50%	70.83%

Notes: All models are estimated via non-linear least squares, and we reports classification results from the balanced sample. The exponential model and single-property models achieve convergence for the same 72 of 79 subjects and the unrestricted model achieves convergence for 63 individuals. All percentages in the table are with respect to the 72 subjects for whom at least once model converges, where we set the AIC weight for the unrestricted model to zero when it does not converge. We implement the small-sample AIC adjustment for every model, adding the penalty term $\frac{2k(k+1)}{N-k-1}$ to each AIC where $N = 55$ is the number of observations per individual, and $k = 1, 2, 4$ is the number of parameters of the model.

8 Discussion

To the degree previous literature considers time-varying time preferences, they are either a confound to the observation of time-inconsistency or an outcome used to assess the causal effect of some shock. In most cases, researchers have relied on at most two points of observation to draw their conclusions. By creating a panel data set with seven observations of discounting over twelve weeks, we offer both a richer environment to study the relationship between time-variance and time-inconsistency (and non-stationarity as well), and an appropriate testing ground for a discounting model that structurally accommodates time-variance, while nesting more familiar discounting models.

Whether we classify subjects structurally or not, with or without a substantial margin of error, we find that the largest single group of subjects by adherence to the TICS properties is none. We also find some evidence of stationary-only behavior that is previously undocumented in the literature, and may help explain why the time preference elicitation literature is so split on whether “present-bias” over monetary rewards is a common phenomenon. Time-variance is more common in our data than time-invariant present-biased discounting, although neither should be labeled as the rule or exception to it; they are both common. This means that the identification of time preferences in traditional time-invariant models may be confounded by preference instability. Time-consistency is the only property with effectively no empirical support.

While we do not argue that this is a general property of discounting, in a canonical experimental context we find that subjects exhibit systematically decreasing patience over the course of our study. Without adjusting for time-variance, this would’ve led us to dramatically overstate the degree of “behavioral” time-inconsistency in our study. 72% of the time-inconsistency we observe in our data is explained by time-variance.

Beyond highlighting this measurement issue in this particular context –we anticipate that the nature and extent of time-variance is highly context-dependent– we developed a new discount function that allows researchers to estimate a model that is fully general with respect to the TICS properties, yet tractable enough to deliver subject-level parameter

estimates in a simple non-linear least squares estimation. This model is easy to pick up and estimate in a longitudinal research study that features all three choices in a decision triangle, but also it can be used in a calibration to explore the robustness of estimates of time-inconsistency or non-stationarity derived from only two of the three choices.

The presence and acknowledgment of time-varying preferences has potential to impact policy design, particularly relating to commitment mechanisms and behavioral nudges. If a policy is meant to reinforce commitment to a consumption plan, take-up of the commitment mechanism may be lower than expected when time-varying preferences exist and are internalized by the policy target. If the target understands that their preferences are changing over time, committing to a previous decision would be welfare-damaging. Similarly, the effect of behavioral nudges may be overstated if the target has time-varying preferences that coincide with the direction the policy is meant to push them.

A deeper exploration of the sources of time-varying behavior in our data is left for future work. Past empirical work on time-variance focuses on risk, uncertainty, and liquidity-constraint as likely causes of this behavior, but learning, anticipation, and anxiety could be others. Because our context is representative of typical experimental elicitations of time preferences in the economics literature, this work could yield methodological insight into whether these elicitations capture transient environmental factors in addition to some time-invariant underlying preferences, or whether preferences are evolving over time as a part of some more general process.

References

- Andersen, S., Harrison, G. W., Lau, M. I., and Rutström, E. E. (2014). Discounting behavior: A reconsideration. *European Economic Review*, 71:15–33.
- Andersen, S., Harrison, G. W., Lau, M. I., and Rutström, E. E. (2008). Eliciting Risk and Time Preferences. *Econometrica*, 76(3):583–618.
- Andreoni, J., Kuhn, M. A., List, J. A., Samek, A., Sokal, K., and Sprenger, C. (2019).

- Toward an understanding of the development of time preferences: Evidence from field experiments. *Journal of Public Economics*, 177:104039.
- Andreoni, J. and Sprenger, C. (2012). Estimating time preferences from convex budgets. *American Economic Review*, 102(7):3333–56.
- Belot, M., Kircher, P., and Muller, P. (2025). Eliciting time preferences when income and consumption vary: Theory, validation, and application to job search. *American Economic Journal: Microeconomics*, 17(1):130–70.
- Bernheim, B. D., Braghieri, L., Martínez-Marquina, A., and Zuckerman, D. (2021). A theory of chosen preferences. *American Economic Review*, 111(2):720–754.
- Bernheim, B. D. and Taubinsky, D. (2018). Behavioral public economics. *Handbook of behavioral economics: Applications and Foundations 1*, 1:381–516.
- Brownback, A., Imas, A., and Kuhn, M. A. (2025). Behavioral food subsidies. *The Review of Economics and Statistics*, 107(3):639–652.
- Burness, H. S. (1976). A note on consistent naive intertemporal decision making and an application to the case of uncertain lifetime. *The Review of Economic Studies*, 43(3):547–549.
- Burnham, K. P. and Anderson, D. R. (2002). *Model selection and multimodel inference: a practical information-theoretic approach*. Springer.
- Burnham, K. P. and Anderson, D. R. (2004). Multimodel inference: Understanding aic and bic in model selection. *Sociological Methods & Research*, 33(2):261–304.
- Carrera, M., Royer, H., Stehr, M., Sydnor, J., and Taubinsky, D. (2022). Who chooses commitment? evidence and welfare implications. *Review of Economic Studies*, 89:1205–1244.

- Carvalho, L. S., Meier, S., and Wang, S. W. (2016). Poverty and economic decision-making: Evidence from changes in financial resources at payday. *American Economic Review*, 106(2):260–84.
- Charness, G., Chemaya, N., and Trujano-Ochoa, D. (2023). Learning your own risk preferences. *Journal of Risk and Uncertainty*, 67:1–19.
- DeJarnette, P. (2020). Temptation over time: Delays help. *Journal of Economic Behavior and Organization*, 177:752–761.
- Drouhin, N. (2020). Non-stationary additive utility and time consistency. *Journal of Mathematical Economics*, 86:1–14.
- Enke, B., Graeber, T., and Oprea, R. (2025). Complexity and time. *Journal of the European Economic Association*, 23(5):1838–1867.
- Gabaix, X. and Laibson, D. (2022). Myopia and discounting. Working paper.
- Halevy, Y. (2015). Time Consistency: Stationarity and Time Invariance. *Econometrica*, 83(1):335–352.
- Harrison, G. W., Lau, M. I., and Yoo, H. I. (2025). Constant discounting, temporal instability, and dynamic inconsistency in denmark: A longitudinal field experiment. *International Economic Review*, 66(1):363–392.
- Hoeting, J. A., Madigan, D., Raftery, A. E., and Volinsky, C. T. (1999). Bayesian model averaging: a tutorial (with comments by m. clyde, david draper and ei george, and a rejoinder by the authors. *Statistical science*, 14(4):382–417.
- Imas, A., Kuhn, M. A., and Mironova, V. (2022). Waiting to choose: The role of deliberation in intertemporal choice. *American Economic Journal: Microeconomics*, 14(3):414–40.

- Jamison, D. T. and Jamison, J. (2011). Characterizing the Amount and Speed of Discounting Procedures. *Journal of Benefit-Cost Analysis*, 2(2):1–56.
- Janssens, W., Kramer, B., and Swart, L. (2017). Be patient when measuring hyperbolic discounting: Stationarity, time consistency and time invariance in a field experiment. *Journal of Development Economics*, 126:77–90.
- Kuhn, M. A. (2025). Eliciting time preferences. Working paper.
- Kuhn, M. A., Kuhn, P., and Villeval, M. C. (2017). Decision-environment effects on intertemporal financial choices: How relevant are resource-depletion models? *Journal of Economic Behavior and Organization*, 137:72–89.
- Laibson, D. (1997). Golden eggs and hyperbolic discounting. *Quarterly Journal of Economics*, 112(2):443–477.
- O’Donoghue, T. and Rabin, M. (1999). Doing it now or later. *American Economic Review*, 89(1):103–124.
- Prelec, D. (2004). Decreasing impatience: A criterion for non-stationary time preference and “hyperbolic” discounting. *The Scandinavian Journal of Economics*, 106(3):511–532.
- Read, D. and van Leeuwen, B. (1998). Predicting hunger: The effects of appetite and delay on choice. *Organizational Behavior and Human Decision Processes*, 76(2):189–205.
- Strotz, R. H. (1956). (1955-1956). myopia and inconsistency in dynamic utility maximization. *The Review of Economic Studies*.-1955-1956, 23(3):165.
- Strulik, H. (2021). Hyperbolic discounting and the time-consistent solution of three canonical environmental problems. *Journal of Public Economic Theory*, 23(3):462–486.

A Appendix for online publication

A.1 Supplementary tables and figures

Table A.1: Aggregate discount parameter estimates from quasi-hyperbolic model, full sample

Model:	OLS	Interval regression	NLS	Binary logistic
	(1)	(2)	(3)	(4)
δ	0.9980 (0.0002)	0.9980 (0.0002)	0.9981 (0.0002)	0.9984 (0.0002)
β	0.9712 (0.0024)	0.9716 (0.0025)	0.9724 (0.0024)	0.9778 (0.0024)
r (annual discount rate)	0.1074 (0.0120)	0.1104 (0.0136)	0.1047 (0.0118)	0.0877 (0.0115)
$H_0: \delta = 1$	$p < 0.0001$	$p < 0.0001$	$p < 0.0001$	$p < 0.0001$
$H_0: \beta = 1$	$p < 0.0001$	$p < 0.0001$	$p < 0.0001$	$p < 0.0001$

Notes: standard errors are clustered at the subject level. δ is the weekly exponential discount factor, and $r = \delta^{-52} - 1$ is the annual discount rate. The data consist of 6,367 price lists from 153 subjects in the fullsample. The binary logistic model assumes a standard T1EV error distribution, and uses data from 140,074 binary choices.

Table A.2: Separate estimates of non-stationarity and inconsistency, full sample

Measure: Dependent variable:	Non-stationarity		Inconsistency	
	d	$\ln(d)$	d	$\ln(d)$
	(1)	(2)	(3)	(4)
Immediate	-0.001 (0.001)	-0.001 (0.001)	-0.004*** (0.001)	-0.005*** (0.002)
Constant (Delay = 8 weeks, Immediate = 0, Week = 0)	0.963 (0.003)	-0.039 (0.004)	0.967 (0.004)	-0.035 (0.004)
Survey-week FE	Y	Y	N	N
Drawing-week FE	N	N	Y	Y
Clusters	153	153	153	153
Observations	6,367	6,367	6,367	6,367

Notes: OLS regressions with standard errors clustered at the subject level. All models include delay-length fixed effects with eight weeks as the omitted group. Week zero is always is omitted group from the survey-week and drawing-week fixed effects. Data are limited to the balanced sample.

Table A.3: Aggregate parameter estimates for the nested exponential model, full sample

Restrictions:	$\mu = \eta = \gamma = 1$	$\mu = \gamma = 1$	$\mu = \eta = 1$	$\eta = \gamma = 1$	None
Properties					
Invariant	✓	✓	X	X	X
Stationary	✓	X	✓	X	X
Consistent	✓	X	X	✓	X
	(1)	(2)	(3)	(4)	(5)
δ	0.9931 (0.0005)	0.9809 (0.0018)	0.9942 (0.0006)	0.9917 (0.0019)	0.9814 (0.0020)
η	1 .	0.5513 (0.0247)	1 .	1 .	0.5958 (0.0462)
γ	1 .	1 .	0.9660 (0.0126)	1 .	0.9605 (0.0179)
μ	1 .	1 .	1 .	0.9400 (0.0756)	0.8481 (0.1132)
AIC	37,812	37,645	37,787	37,813	37,625
AIC weight	< 0.0001	0.0001	< 0.0001	< 0.0001	0.9999
Bayesian PMP	< 0.0001	0.0001	< 0.0001	< 0.0001	0.9999
r_0	0.4322 (0.0401)	0.1852 (0.0182)	0.3498 (0.0404)	0.4064 (0.0534)	0.1480 (0.0207)
r_{12}	$= r_0$	$= r_0$	0.5749 (0.0871)	0.3890 (0.0688)	0.2348 (0.0357)
PB_S	1 .	0.9799 (0.0020)	1 .	0.9978 (0.0030)	0.9985 (0.0110)
PB_C	1 .	$= PB_S$	0.9967 (0.0011)	1 .	0.9791 (0.0018)
$H_0: \delta = 1$	$p < 0.0001$	$p < 0.0001$	$p < 0.0001$	$p < 0.0001$	$p < 0.0001$
$H_0: \eta = 1$	.	$p < 0.0001$	.	.	$p < 0.0001$
$H_0: \gamma = 1$	.	.	$p = 0.0076$	.	$p = 0.0287$
$H_0: \mu = 1$	.	.	.	$p = 0.4284$	$p = 0.1817$
$H_0: \mu = \gamma = 1$	.	.	.	.	$p = 0.0766$
$H_0: \mu = \eta = 1$	.	.	.	.	$p < 0.0001$
$H_0: \eta = \gamma = 1$	.	.	.	.	$p < 0.0001$
$H_0: \mu = \eta = \gamma = 1$	.	.	.	.	$p < 0.0001$
$H_0: r_0 = r_{12}$	.	.	$p = 0.0183$	$p = 0.3954$	$p = 0.0317$
$H_0: PB_S = 1$	.	$p < 0.0001$	.	$p = 0.4578$	$p = 0.8925$
$H_0: PB_C = 1$	.	$p = 0.0029$	$p = 0.0200$	.	$p < 0.0001$
$H_0: PB_S = PB_C$	.	.	.	.	$p = 0.0833$

Notes: Standard errors are clustered at the subject level. $r_\tau = D(\tau, \tau + 52)^{-1} - 1$ is a measure of the annual discount rate. PB_S and PB_C are stationarity-based and consistency-based measures of present-bias. The data consist of 4,659 price lists from 153 subjects, using only the choice sets that spanned every survey. All estimates are from non-linear least squares regressions.

	Prize Draw today	or	Prize Draw in 2 weeks
Choice 1:	202		242
Choice 2:	204		242
Choice 3:	206		242
Choice 4:	208		242
Choice 5:	210		242
Choice 6:	212		242
Choice 7:	214		242
Choice 8:	216		242
Choice 9:	218		242
Choice 10:	220		242
I prefer 222 tickets today to 242 tickets in 2 weeks			
Choice 11:	222		242
Choice 12:	224		242
Choice 13:	226		242
Choice 14:	228		242
Choice 15:	230		242
Choice 16:	232		242
Choice 17:	234		242
Choice 18:	236		242
Choice 19:	238		242
Choice 20:	240		242
Choice 21:	242		242
Choice 22:	244		242
	Prize Draw today	or	Prize Draw in 2 weeks
		Next	

Figure A.1: Example price list

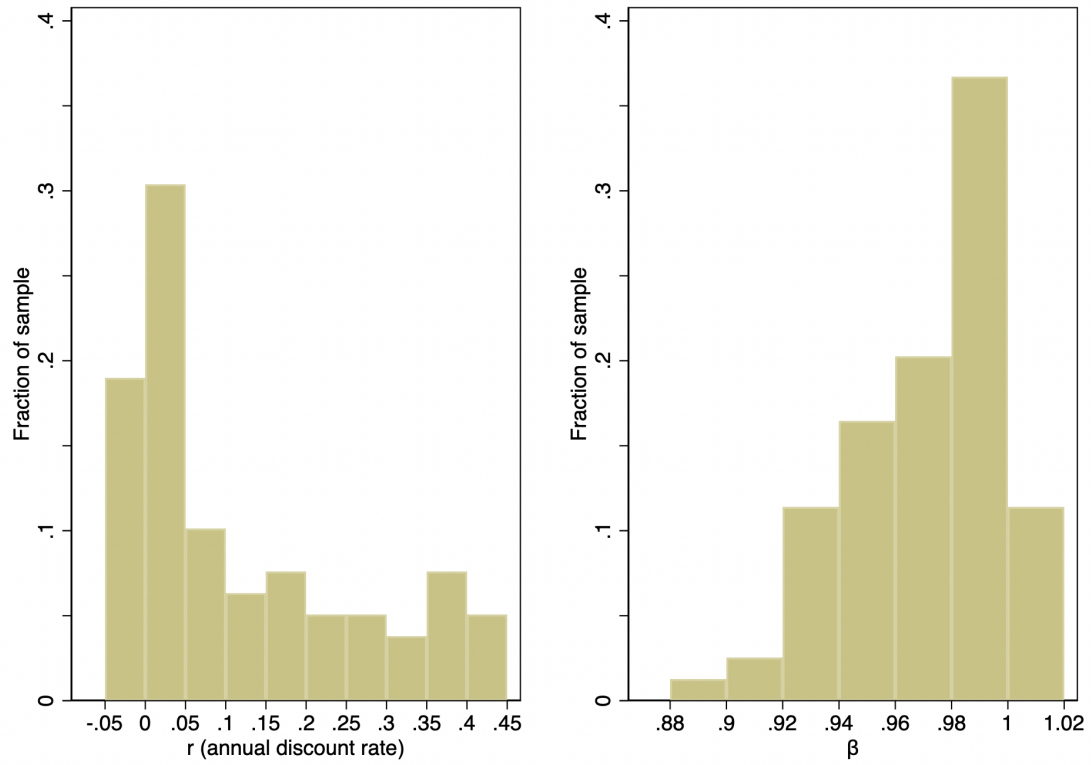


Figure A.2: Distributions of individual estimates from the β - δ model

Notes: β and δ parameters are estimated using the OLS specification given in equation 6. $r = \delta^{-52} - 1$ is the annual discount rate. The data consist of 4,345 price lists from the 79 subjects in the balanced sample.

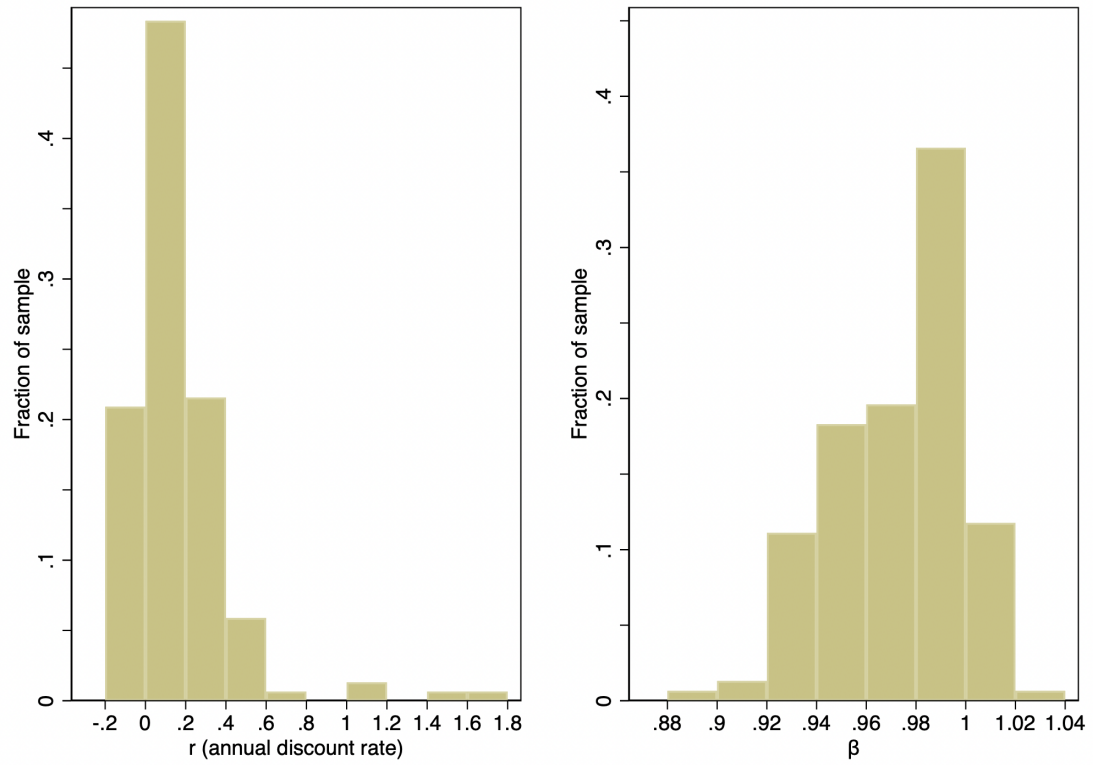


Figure A.3: Distributions of individual estimates from the β - δ model, full sample

Notes: β and δ parameters are estimated using the OLS specification given in equation 6. $r = \delta^{-52} - 1$ is the annual discount rate. The data consist of 6,367 price lists from the 153 subjects in the full sample.

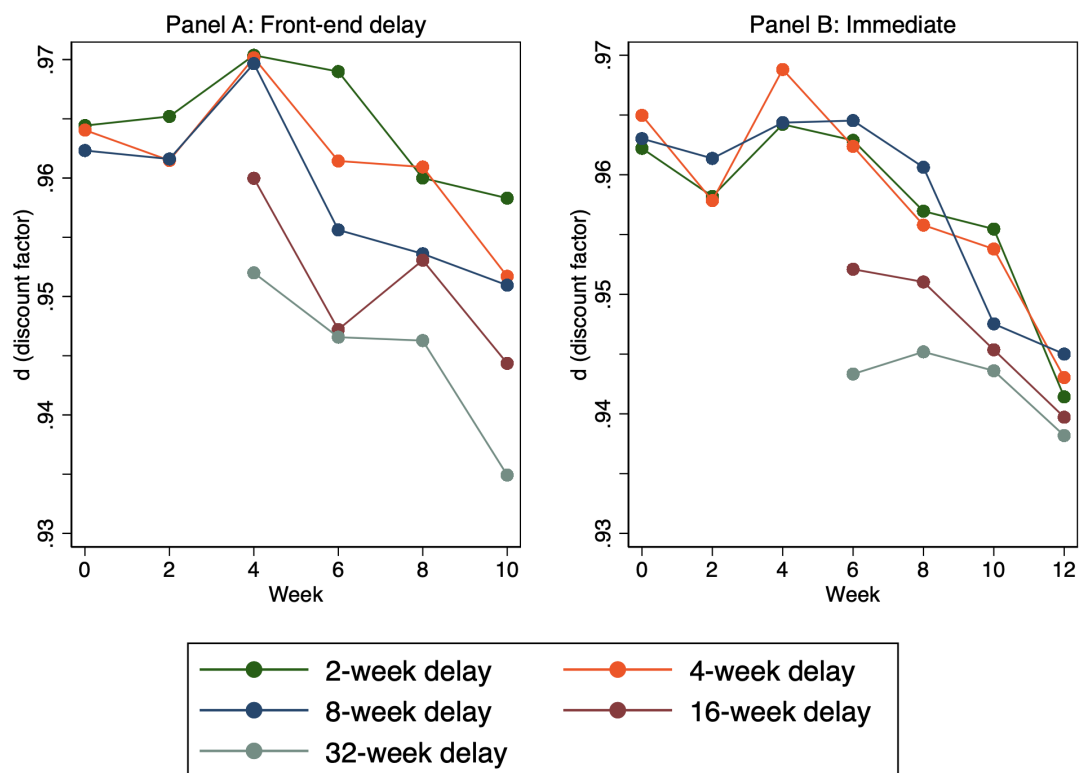


Figure A.4: Time variance in discounting, full sample

Notes: The data consist of 6,367 price lists from the 153 subjects in the full sample.

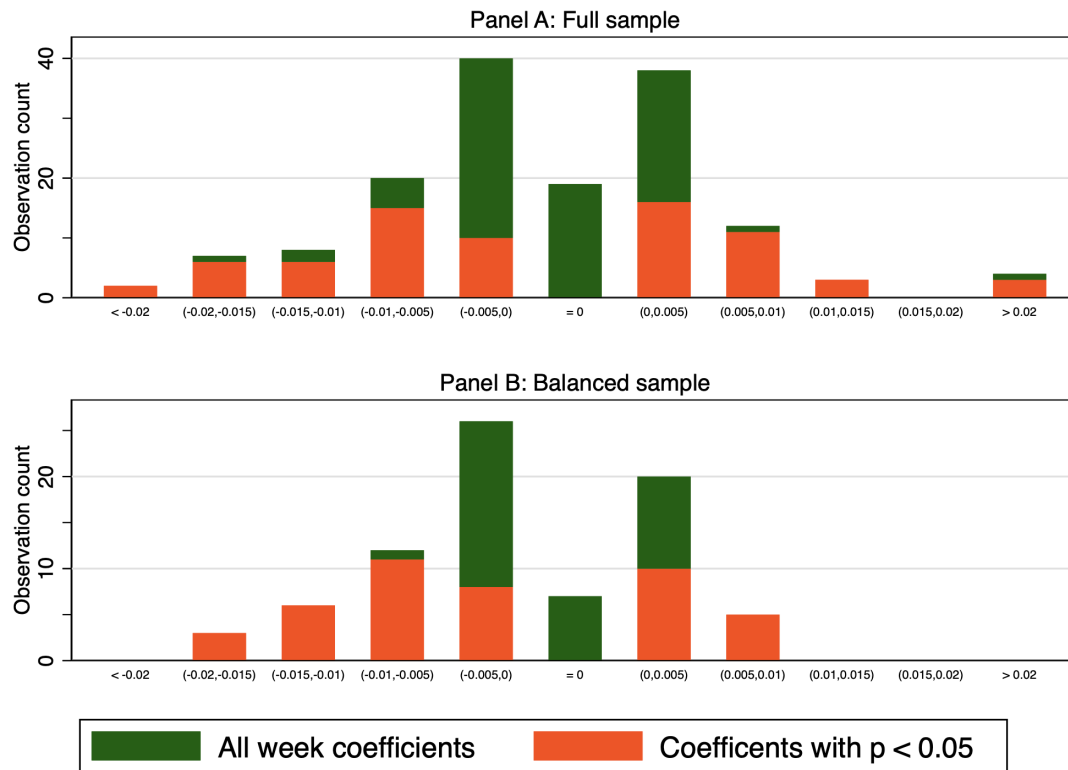


Figure A.5: Subject-level estimates of time variance

Notes: Coefficient estimates are from OLS regressions of the discount factor (d) on a survey week variable with fully-interacted delay length and front-end delay fixed effects at the subject-level with heteroskedasticity-robust standard errors. There are 153 subjects in the full sample and 79 subjects in the balanced sample.

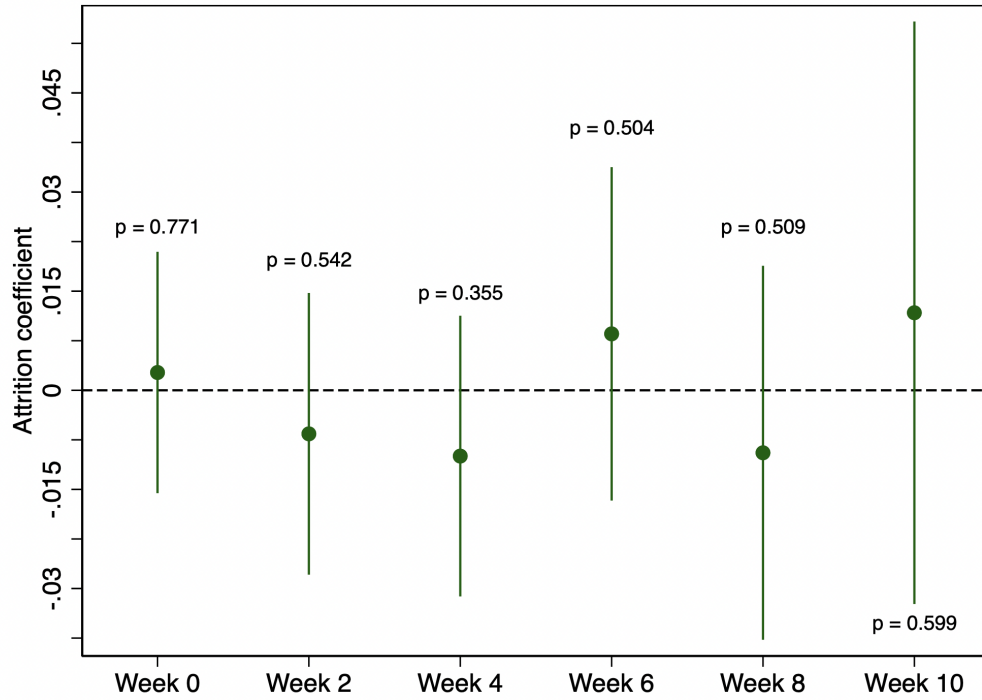


Figure A.6: Relationship between subsequent attrition and discount factor, with 95% confidence intervals

Notes: Coefficients and 95% confidence intervals are from a regression of the discount factor (d) on an indicator for whether a subject fails to complete the subsequent survey. Standard errors are clustered at the subject level. All models feature fully-interacted fixed effects between delay length and front-end delay. Cluster (observation) counts are 153, 138, 117, 115, 106, and 99 (918, 828, 936, 1,150, 1,060, and 990), respectively.

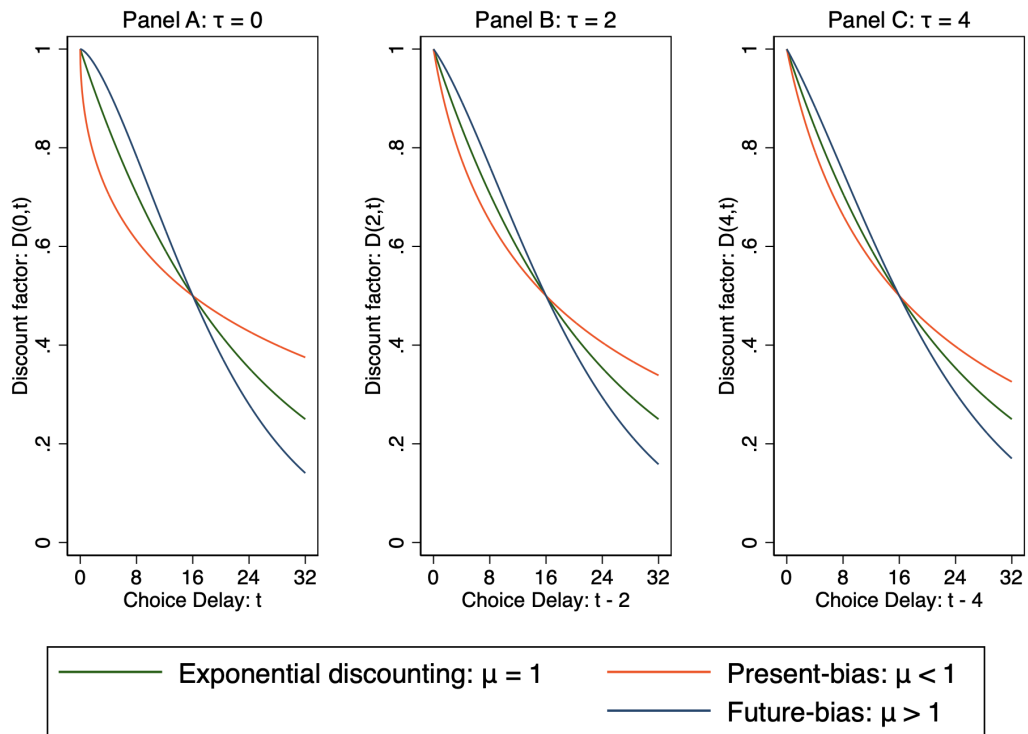


Figure A.7: Time-consistent NED function ($\eta = \gamma = 1$)

Notes: the value of δ in each function is chosen such that $D(\tau, \tau + 16) = 0.5$. We show $\mu = 0.5$ and $\mu = 1.5$.

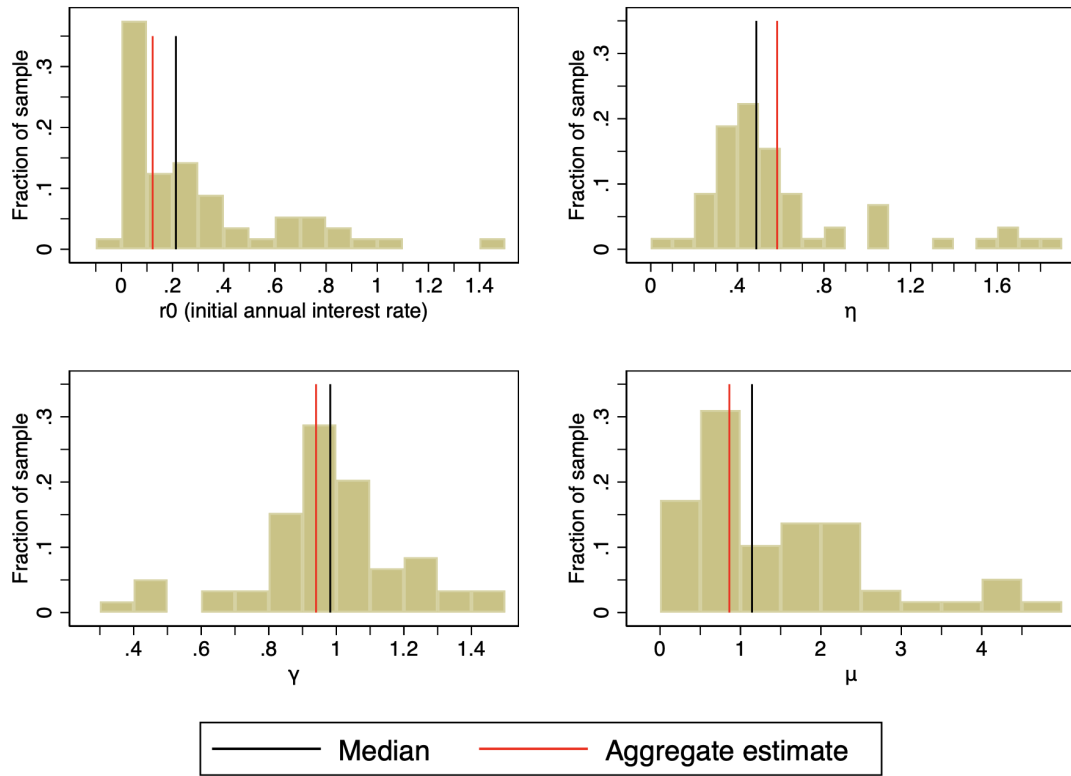


Figure A.8: Individual parameter estimates for the NED function

Notes: Three positive outliers are trimmed from the r_0 distribution, one from the η distribution, and one from the μ distribution for visual clarity, but the median is taken with respect full sets of estimates.

A.2 Proof of Proposition II

Proof. ($\Leftarrow$) Suppose a discount function $D(\tau, t)$ is only a function of $t - \tau$. Given bundles $(\varepsilon, t + \Delta_1) \sim_t (\psi, t + \Delta_2)$ which the agent is indifferent between at evaluation time t , we have

$$\begin{aligned} D(t, t + \Delta_1) \varepsilon &= D(t, t + \Delta_2) \psi \\ \implies D(t + \Delta_1 - t) \varepsilon &= D(t + \Delta_2 - t) \psi \\ \implies D(\Delta_1) \varepsilon &= D(\Delta_2) \psi. \end{aligned}$$

For any t' , we can write $\Delta_i = t' + \Delta_i - t'$ for $i = 1, 2$. The above equation implies

$$D(t' + \Delta_1 - t') \varepsilon = D(t' + \Delta_2 - t') \psi.$$

By assumption, this is equivalent to

$$D(t', t' + \Delta_1) \varepsilon = D(t', t' + \Delta_2) \psi$$

which establishes time-invariance.

($\Leftarrow$)

($\Rightarrow$) Let $D(\tau, t)$ be a time-invariant function. Suppose there exists bundles (ε, τ) and $(\psi, \tau + \Delta)$ which the agent is indifferent between at evaluation time τ . That is,

$$D(\tau, \tau) \varepsilon = D(\tau, \tau + \Delta) \psi \quad .$$

According to the definition of time-invariance, the agent's discounted utility must satisfy

$$D(\tau + t, \tau + t) \varepsilon = D(\tau + t, \tau + \Delta + t) \psi \quad \forall t \quad .$$

Together, these two equations imply

$$\frac{D(\tau + t, \tau + \Delta + t)}{D(\tau + t, \tau + t)} = \frac{\varepsilon}{\psi} = \frac{D(\tau, \tau + \Delta)}{D(\tau, \tau)} \quad \forall t \quad .$$

We assume in general that agents do not discount the present period, so that $D(x, x) \equiv 1$.

Thus,

$$D(\tau, \tau + \Delta) = D(\tau + t, \tau + \Delta + t) \quad \forall t \quad .$$

Note that Δ is the distance between a payoff time and an evaluation time. For conciseness, assume $\tau = 0$. Then

$$D(0, \Delta) = D(t, \Delta + t) \quad \forall t \quad .$$

($\implies$)

■

A.3 Proof of Proposition IV

Proof. ($\Leftarrow$) We first verify that $D(\tau, t) = \alpha^{(t-\tau)f(\tau)}$ is a stationary function. Consider bundles (ε, t_1) and (ψ, t_2) which the DM is indifferent between at time τ . Consider $\Delta > 0$. Exponent rules tell us

$$\begin{aligned} D(\tau, t_1 + \Delta) \varepsilon &= \alpha^{(t_1 + \Delta - \tau)f(\tau)} \varepsilon \\ &= \alpha^{\Delta f(\tau)} \alpha^{(t_1 - \tau)f(\tau)} \varepsilon \\ &= \alpha^{\Delta f(\tau)} D(\tau, t_1) \varepsilon \quad . \end{aligned}$$

Indifference between (ε, t_1) and (ψ, t_2) tells us $D(\tau, t_1) \varepsilon = D(\tau, t_2) \psi$. Continuing from the previous expression,

$$= \alpha^{\Delta f(\tau)} D(\tau, t_2) \psi \quad .$$

Expanding upon the particular form of the discount function, we can refactor $\alpha^{\Delta f(\tau)} \alpha^{(t_2 - \tau)f(\tau)} \psi$ into $D(\tau, t_2 + \Delta) \psi$. This establishes $D(\tau, t_1 + \Delta) \varepsilon = D(\tau, t_2 + \Delta) \psi$.

($\Leftarrow$)

($\Rightarrow$) Suppose $D(\tau, t)$ admits stationary preferences. Given bundles (ε, t) and $(\psi, t + \Delta)$ which the DM is indifferent between at time τ ,

$$D(\tau, t) \varepsilon = D(\tau, t + \Delta) \psi \quad .$$

Taking the log of both sides, we obtain

$$\log [D(\tau, t)] + \log \varepsilon = \log [D(\tau, t + \Delta)] + \log \psi \quad .$$

Stationarity implies this equality holds for any t . Therefore, we can take the t -derivative on both sides. Let $\dot{D}$ denote the derivative of D with respect to payoff time, t . Then

$$\frac{\dot{D}(\tau, t)}{D(\tau, t)} = \frac{\dot{D}(\tau, t + \Delta)}{D(\tau, t + \Delta)} \quad .$$

Therefore, the rate of change of the discount function is independent of payoff time. In other words, for some function $g(\cdot)$, stationarity implies

$$\frac{\dot{D}(\tau, t)}{D(\tau, t)} \equiv g(\tau).$$

One solution to this differential equation is

$$D(\tau, t) = a \cdot e^{f_1(\tau) \cdot t + f_2(\tau)}$$

where $f'_1 = g$ and a is a constant.²⁰ Since we require $D(0, 0) = 1$, we can say that $a = 1$. Furthermore, we require that $D(\tau, t) = 1$ whenever $\tau = t$, restricting our exponent to satisfy

$$\begin{aligned} f_1(\tau) \cdot \tau + f_2(\tau) &= 0 \\ \implies f_2(\tau) &= -\tau \cdot f_1(\tau). \end{aligned}$$

Consequently, we can drop the subscripts on f and our exponent can be reduced to $tf(\tau) - \tau f(\tau) = (t - \tau)f(\tau)$. That is, our stationary discount function is given by

$$D(\tau, t) = e^{(t-\tau)f(\tau)}.$$

To allow for full generality, we replace e with some $\alpha > 0$.²¹ Our function then takes the final general form

$$D(\tau, t) = \alpha^{(t-\tau)f(\tau)}.$$

($\implies$)

■

²⁰ $h(x) = a \cdot e^{kx+b}$ solves the differential equation $h'(x) = k \cdot h(x)$. The extension to our multivariate model is differential equation: $\dot{D}(\tau, t) = g(\tau)D(\tau, t)$, where $g(\tau)$ is constant with respect to t .

²¹The arbitrary choice of a is now, more properly, $(\ln a)^{-1}$.